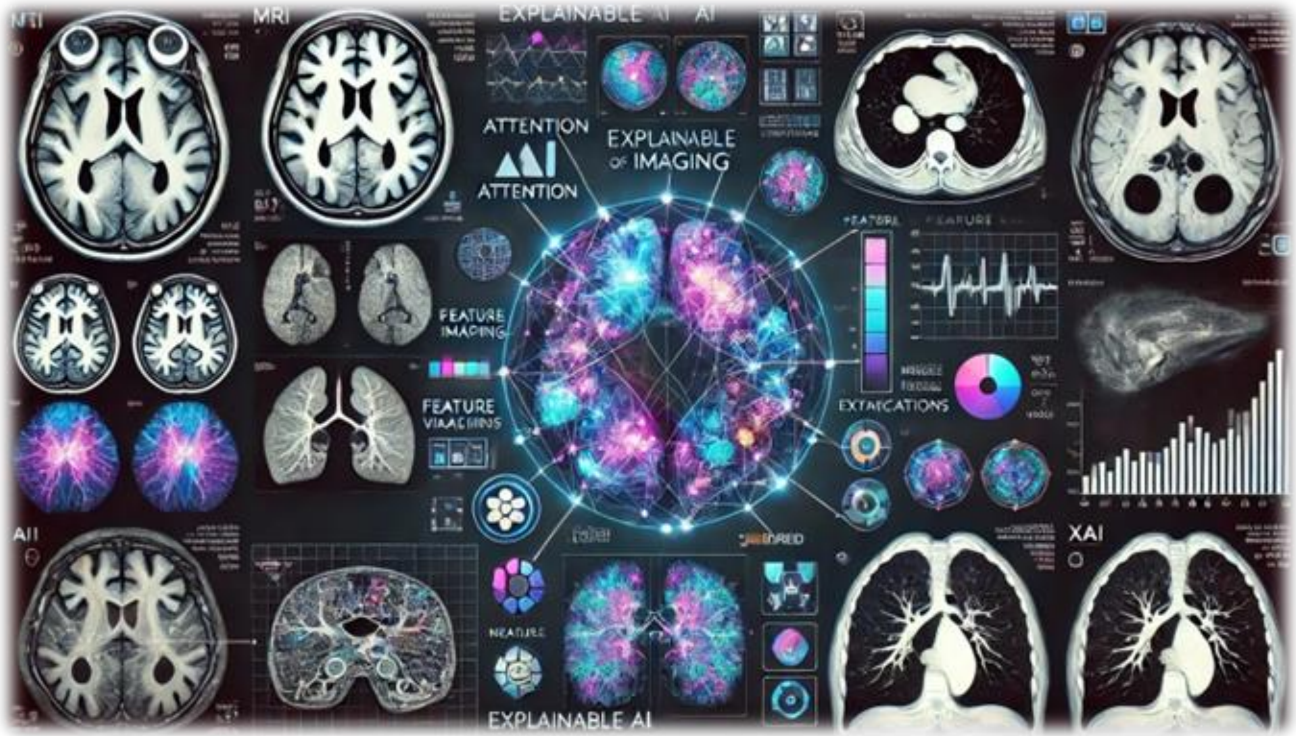
XAI in Medical Image Analysis

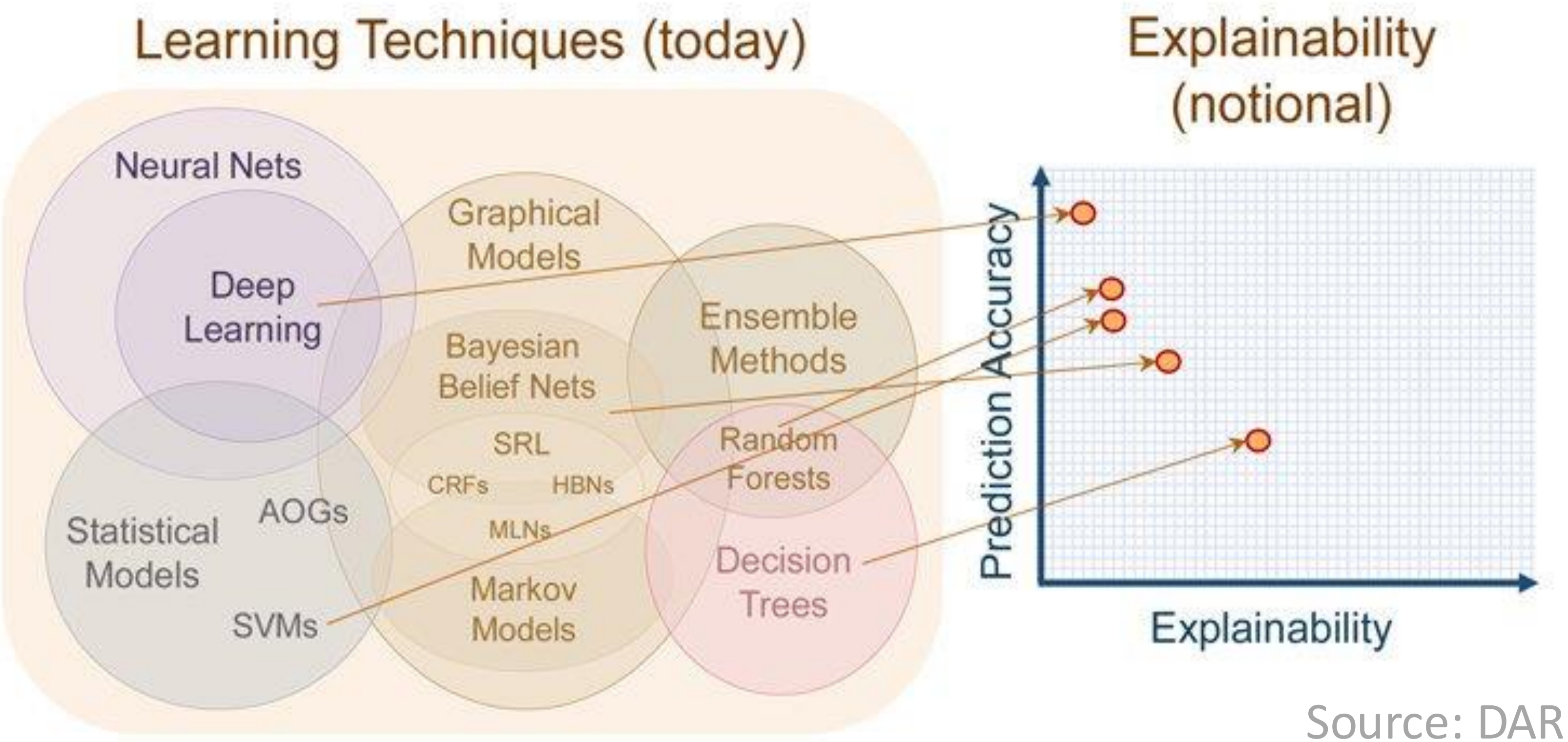


Mauricio Reyes, PhD.

ETH zürich

UNIVERSITÄT BERN

Explainable A.I



Source: DARPA

Definitions

- What is Interpretability?

An interpretable machine learning algorithm is described as one where the **link between the features** used by the machine learning and the **prediction** itself **can be understood by a human**

Another definition

- Produce explanations <u>without</u> sacrificing accuracy
  - Simpler is easier to understand
  - But oversimplified is typically from an accuracy point of view, not interesting

Barocas, S., Friedler, S., Hardt, M., Kroll, J., Venka-Tasubramanian, S., & H. Wallach, H. (2018). The FAT-ML Workshop Series on Fairness, Accountability, and Transparency in Machine Learning.

nature machine intelligence

PERSPECTIVE

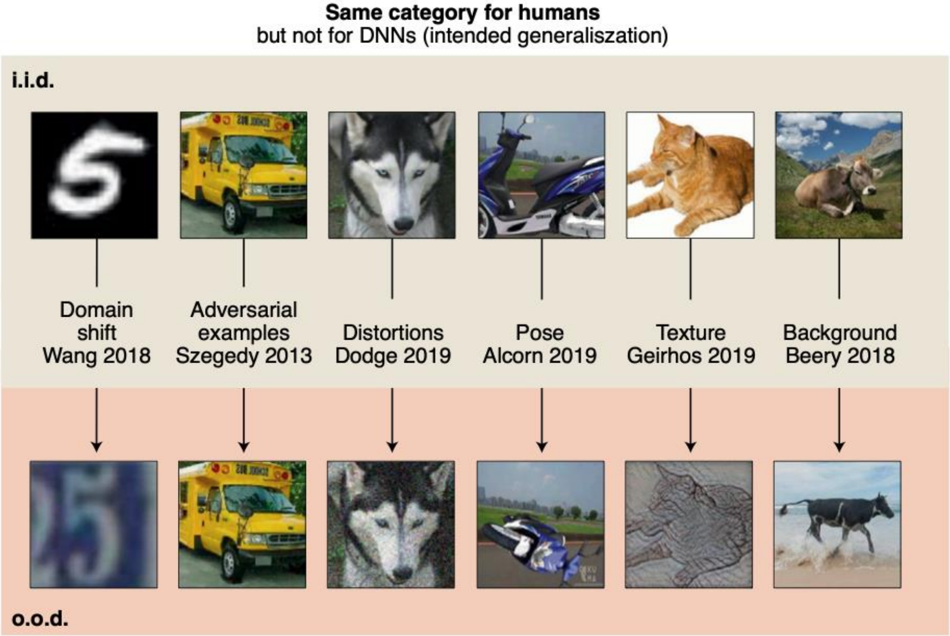
https://doi.org/10.1038/s42256-020-00257-z

Check for updates

Shortcut learning in deep neural networks

Robert Geirhos[1,2,4], Jörn-Henrik Jacobsen[3,4], Claudio Michaelis[1,2,4], Richard Zemel[3,5], Wieland Brendel[1,5], Matthias Bethge[1,5] and Felix A. Wichmann[1,5]

- Principle of “least effort”
- Inductive bias:
  - Model architecture
  - Loss
  - Optimization
  - Training data



nature machine intelligence

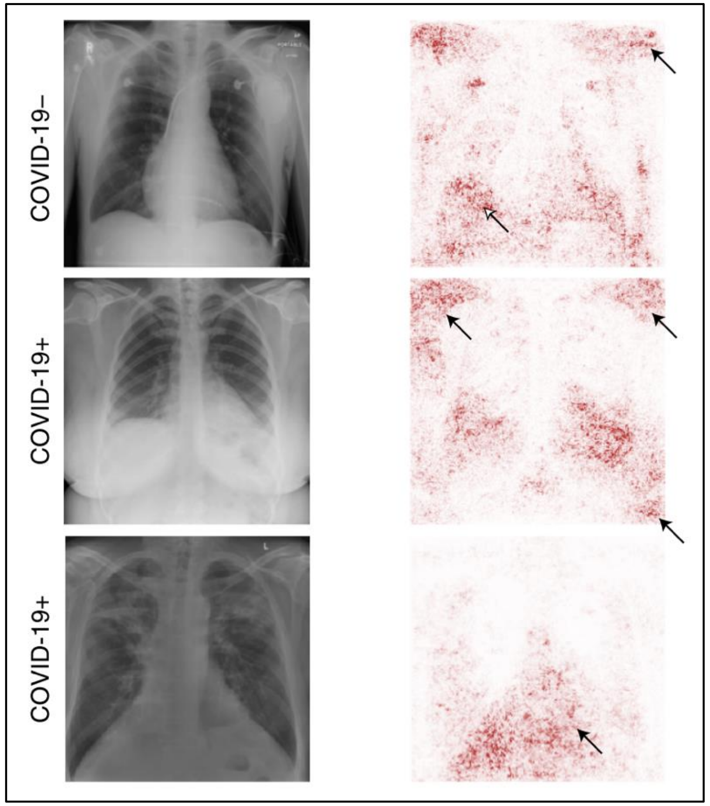
ARTICLES

https://doi.org/10.1038/s42256-021-00338-7

Check for updates

AI for radiographic COVID-19 detection selects shortcuts over signal

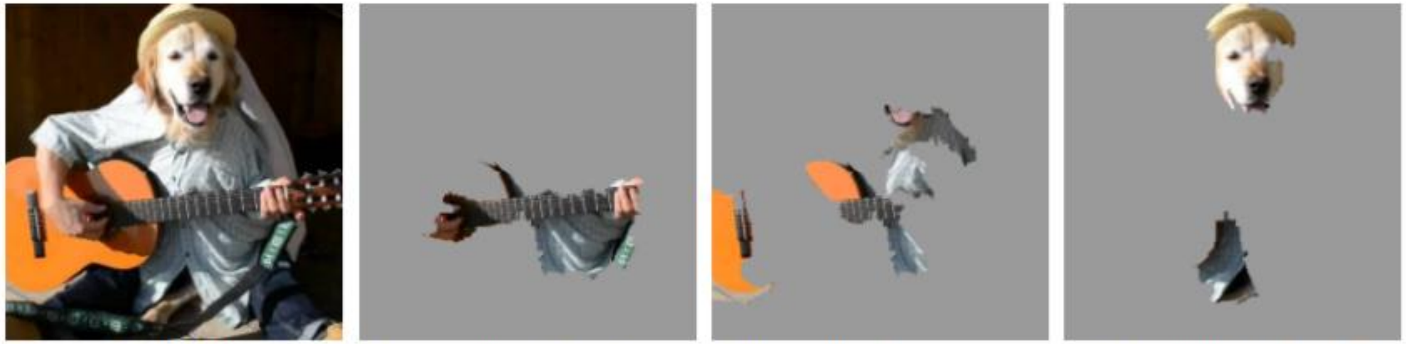
Alex J. DeGrave[1,2,3], Joseph D. Janizek[1,2,3] and Su-In Lee[1]



Laterality markers

Attention out of the region of interest

Explaining DL decisions via attention maps

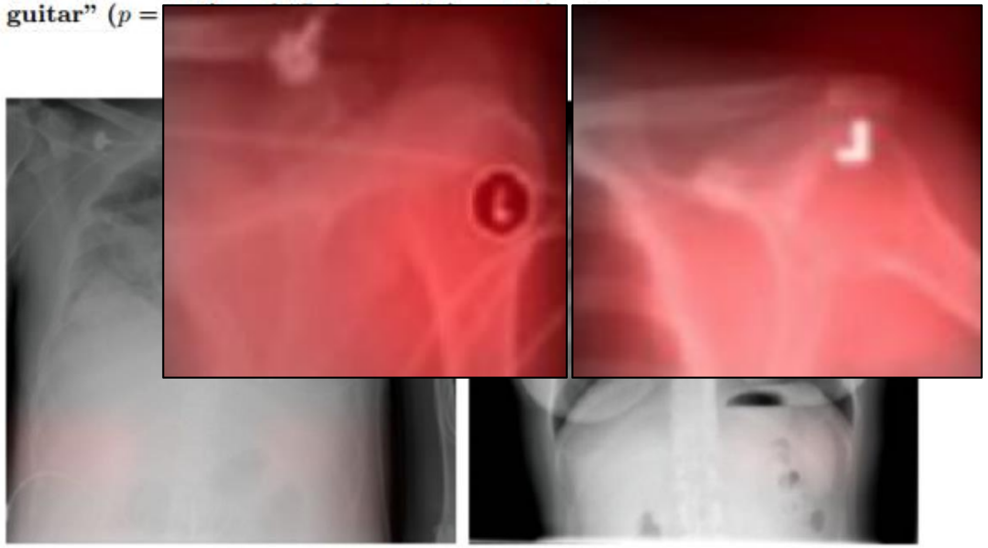


(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

Figure 4: Explaining an image classification prediction made by Google's Inception neural network. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$), "Acoustic guitar" ($p =$

(a) Husky classified as wolf (b) Explanation

Figure 11: Raw data and explanation of a bad model's prediction in the "Husky vs Wolf" task.



DL learned to detect clinical site!

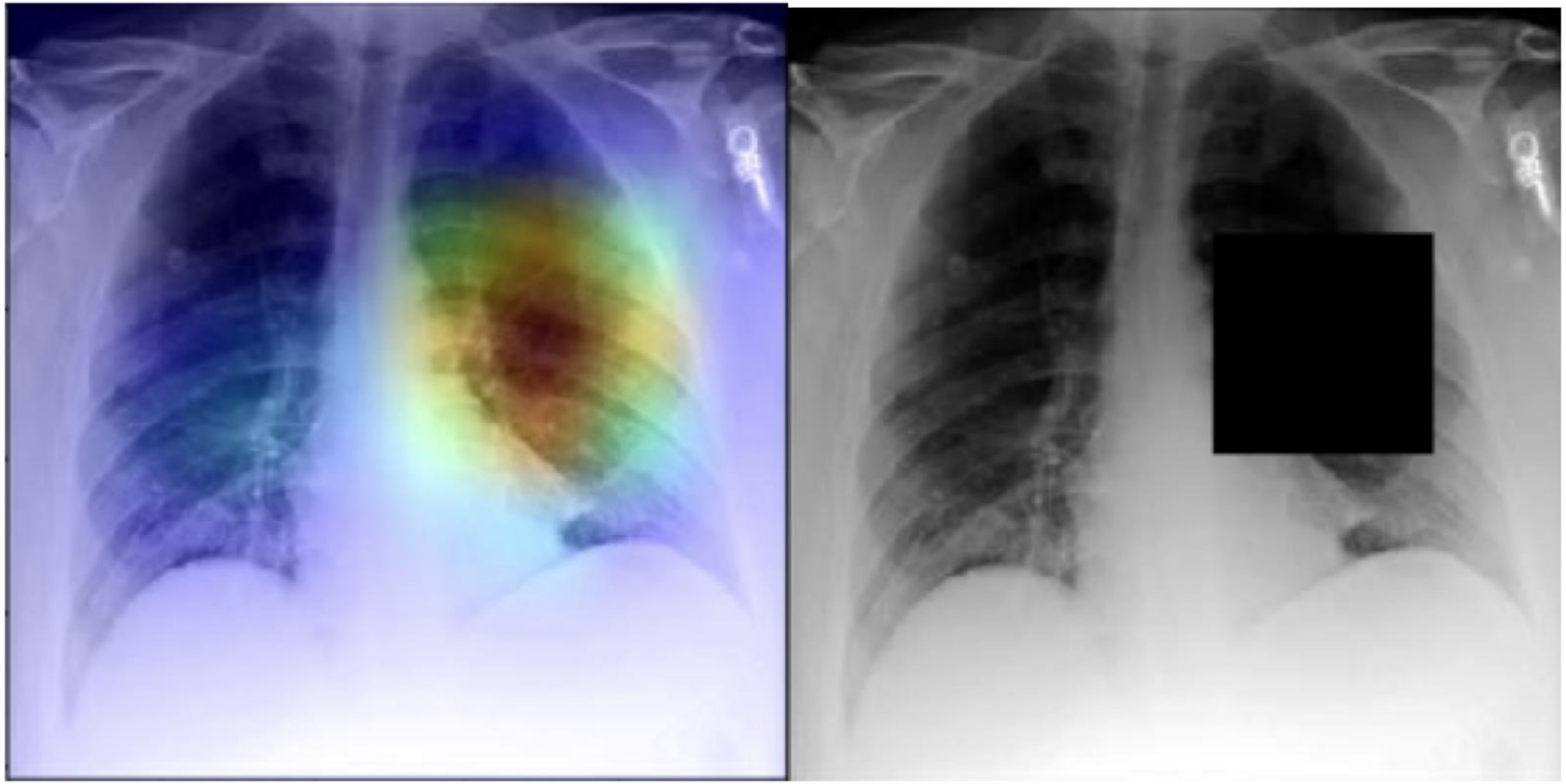
Source: Ribeiro et al. 2017, Zech et al. 2018

AI recognition of patient race in medical imaging: a modelling study

Judy Wawira Gi
Marzyeh Ghas

MXR Dense	(0.930-
MXR Dense	(0.823-

- Detect
- Pattern

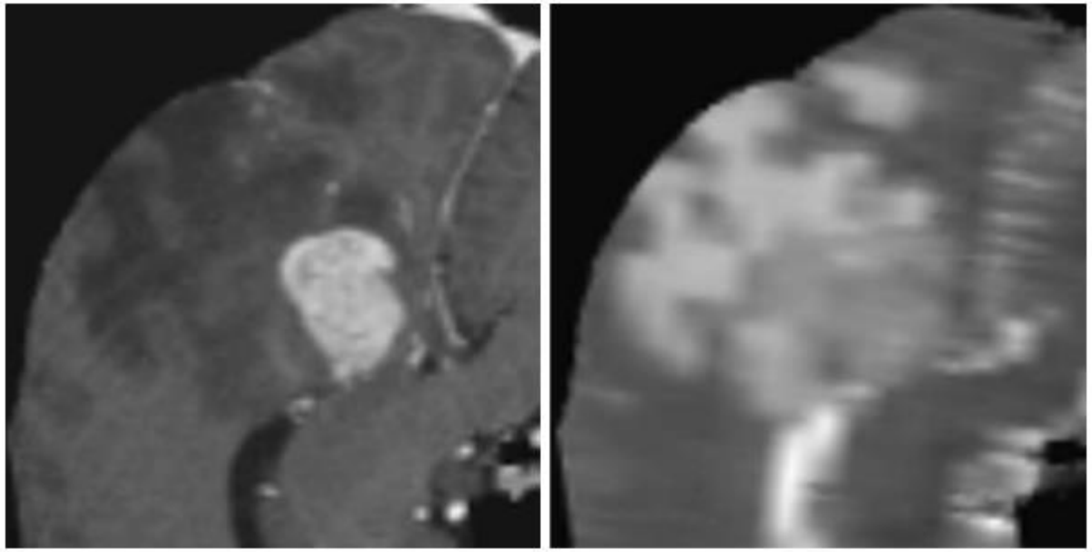


Interpretability in AI Medical Imaging

Automated classification of Low and High-grade gliomas (LGG vs. HGG)

Q: Are there biases stemming from the data preparation process?

A: Bias of learned patterns detected via interpretability



T1c T2

Pereira et al. MICCAI-iMIMIC 2018
Reyes et al. Radiology: Artificial Intelligence 2021

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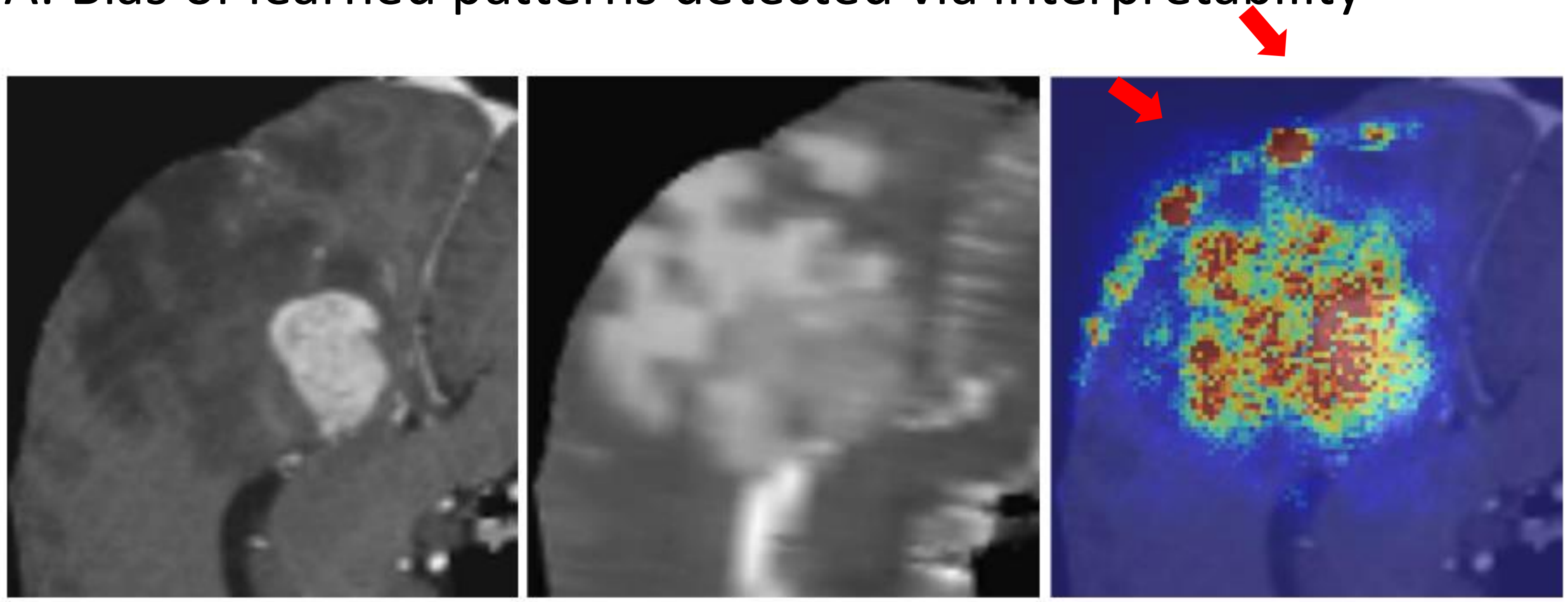

T1c T2

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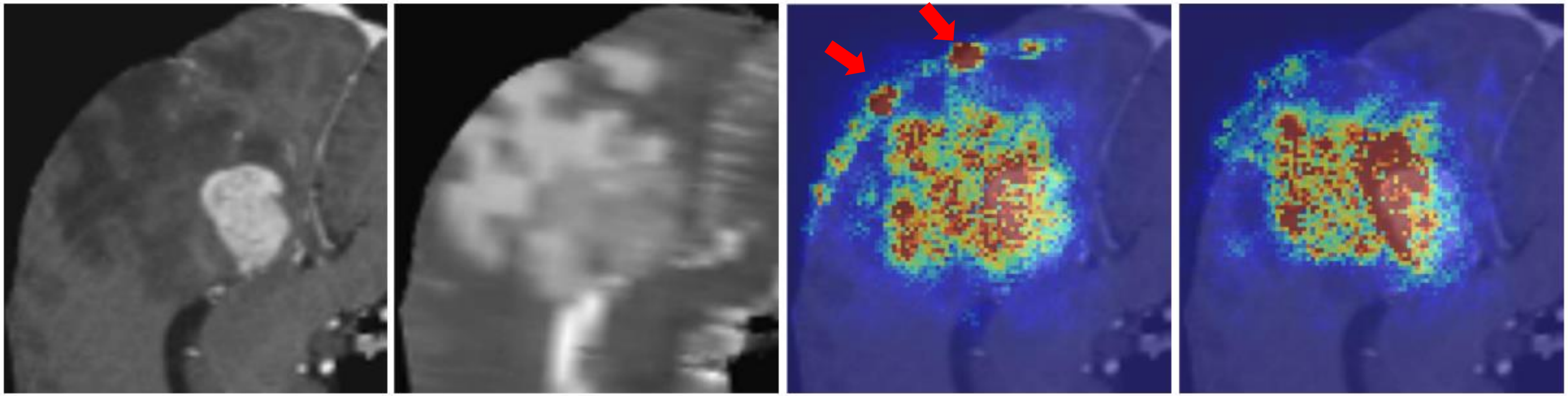
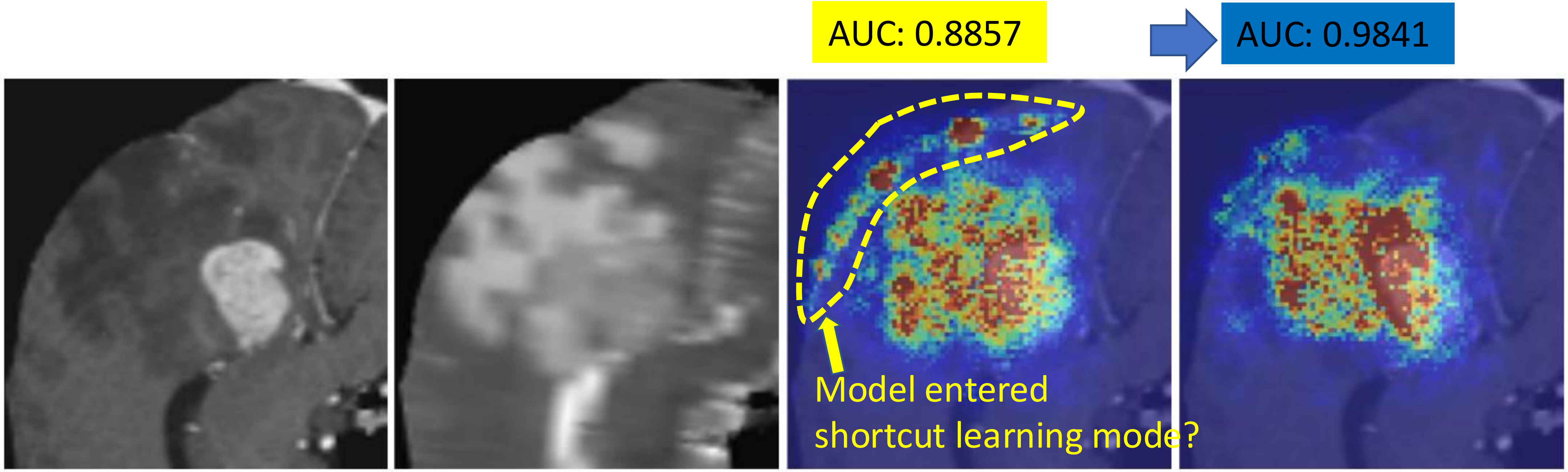


T1c T2 Before After

Saliency Maps

Interpretability in AI Medical Imaging

Key finding: interpretability enhances data preparation and AI-performance



T1c-weighted

T2-weighted

Saliency map:
Normalization
Whole image

Saliency map:
Normalization
Brain area

Pereira et al. MICCAI-iMIMIC 2018
Reyes et al. Radiology: Artificial Intelligence 2021

Published as a conference paper at ICLR 2019

IMAGENET-TRAINED CNNS ARE BIASED TOWARDS TEXTURE; INCREASING SHAPE BIAS IMPROVES ACCURACY AND ROBUSTNESS

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original greyscale silhouette edges texture

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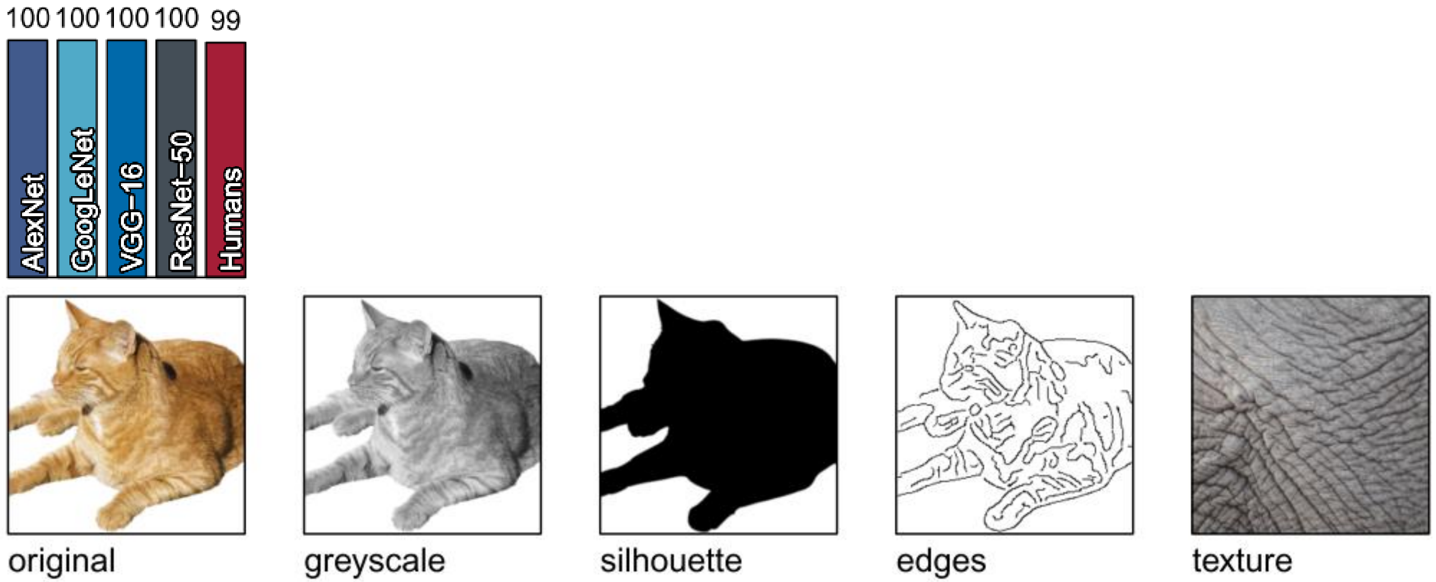

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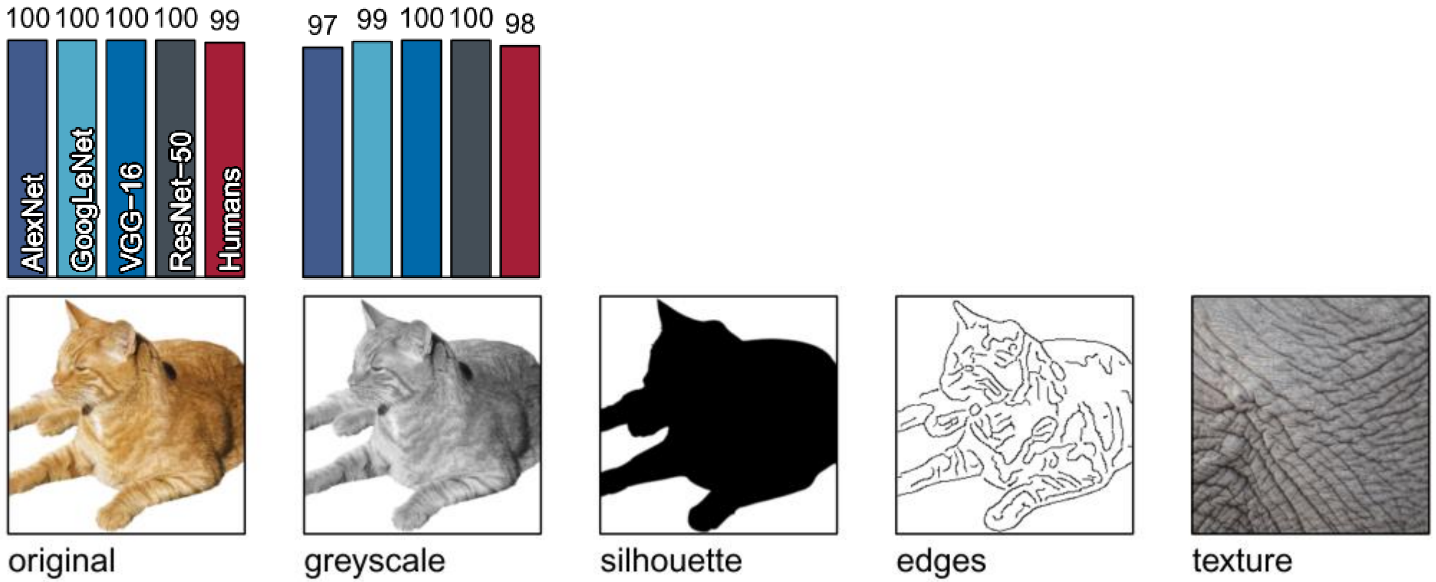

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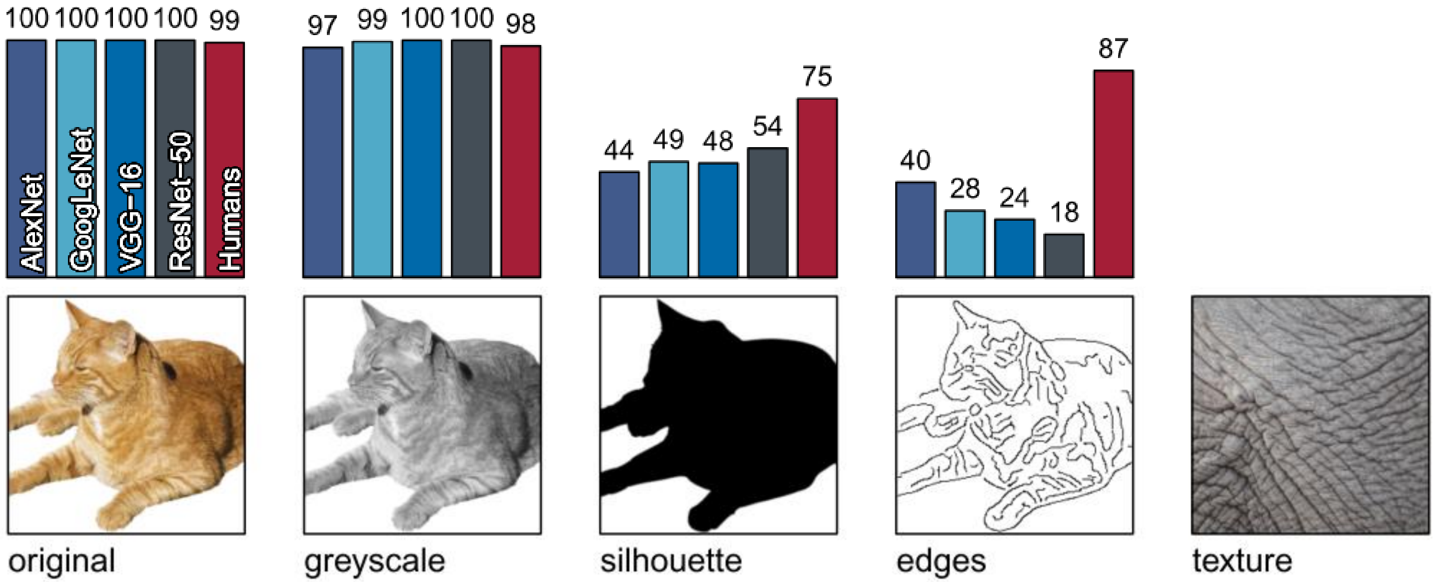

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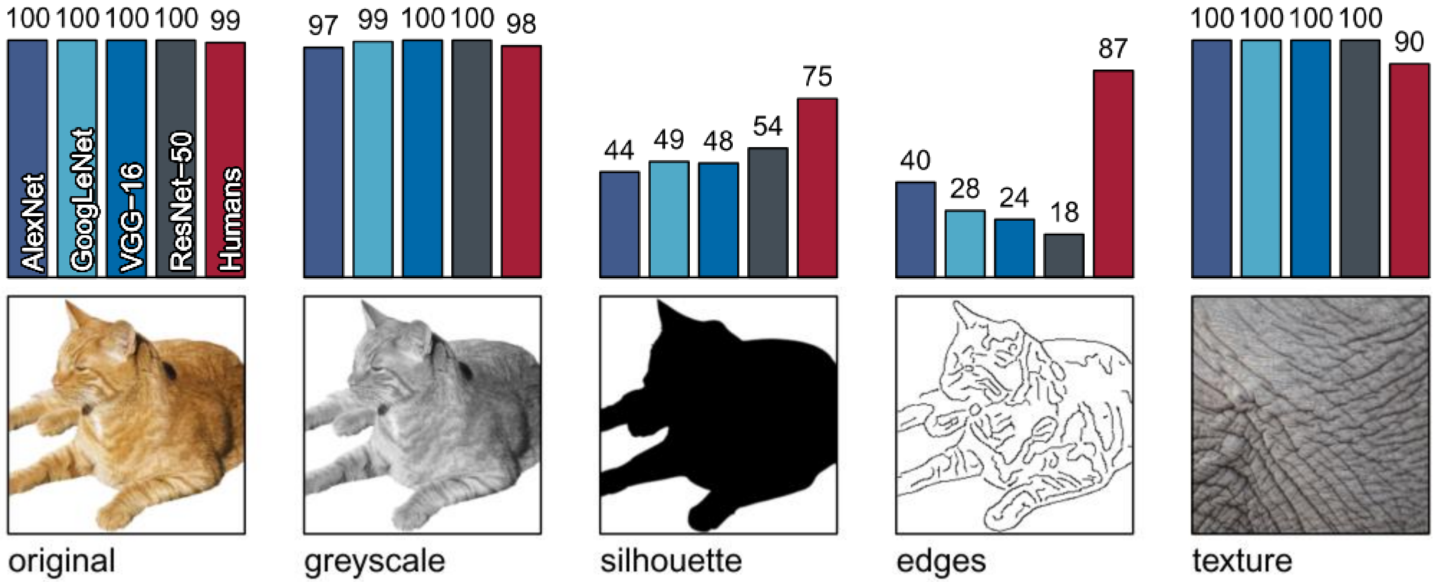


Taxonomy

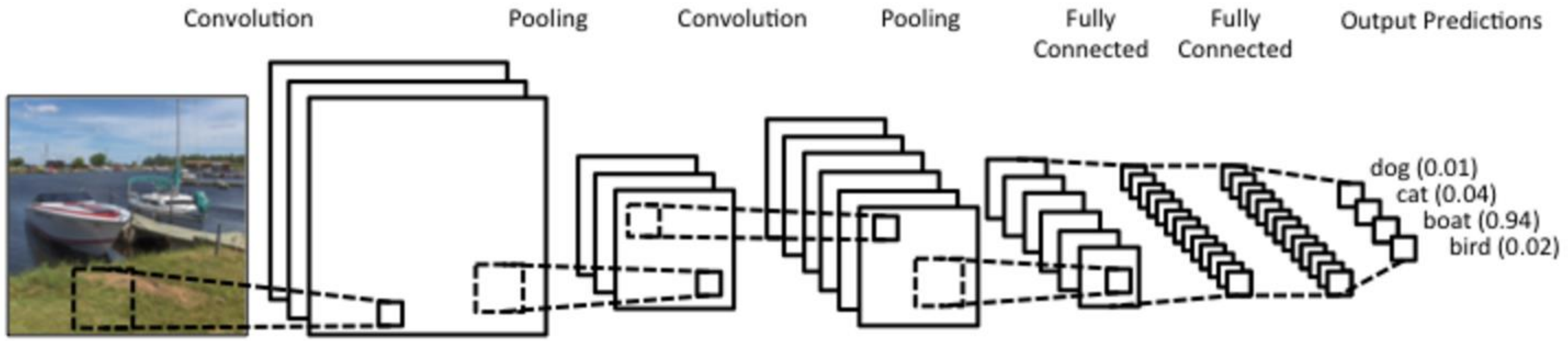
- **Black-boxes** operating methods: No need to have "access" to the internal of models. Also referred to as model-agnostic.
- **White-boxes**: These are methods that require access to the internals of the models. Gradient-based models are one example of white box interpretability models.
- By **output**:
  - Visualization, or saliency maps: Provide pixel-wise values reflecting their importance to the performance of the model being interpreted
  - Concepts: summary statement or keyword.
  - Feature importance: explain models by expressing importance of features E.g. high weights of a model.

Interpretable DL

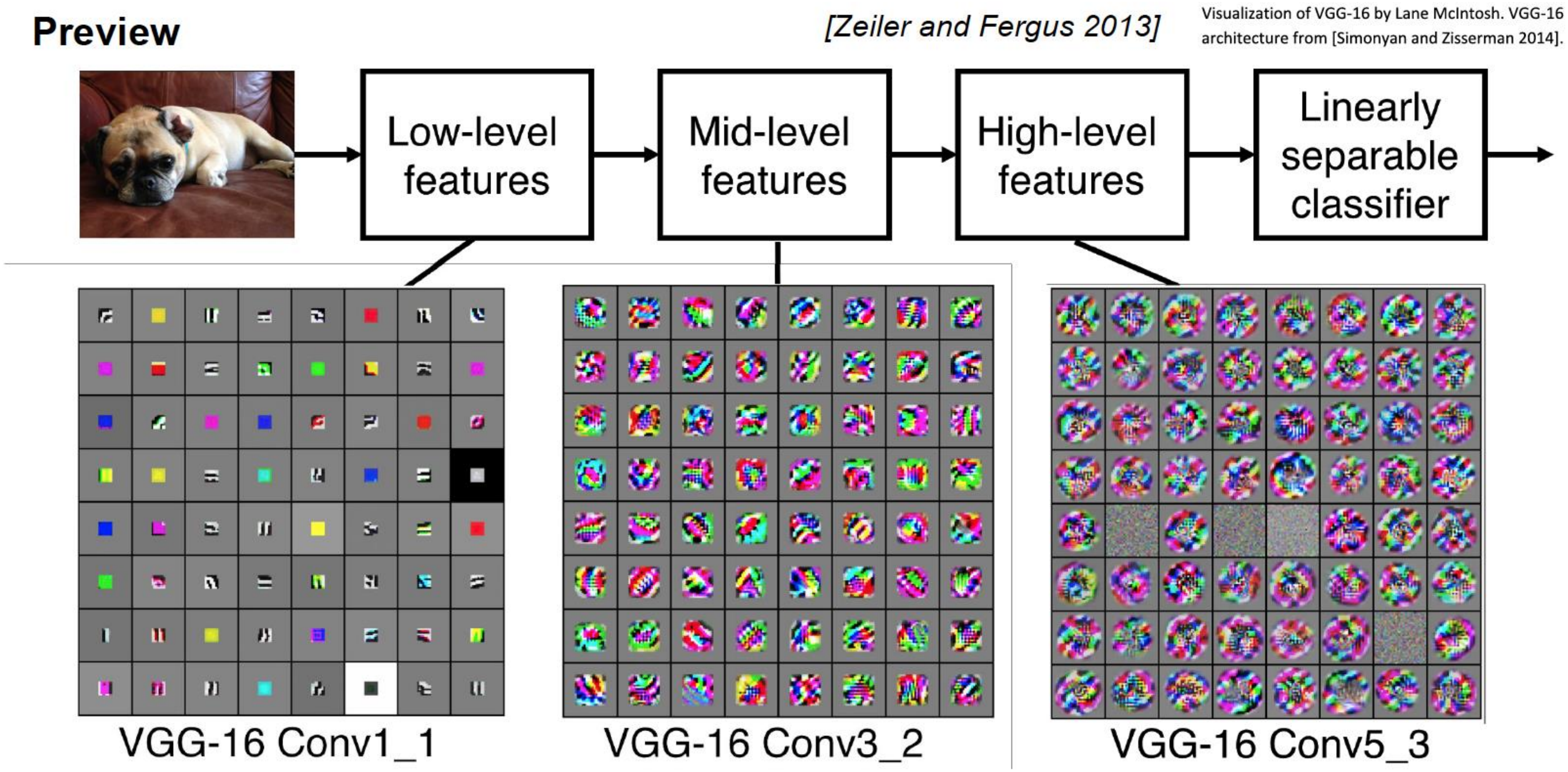
Convolutional Neural Networks

Typical structure:

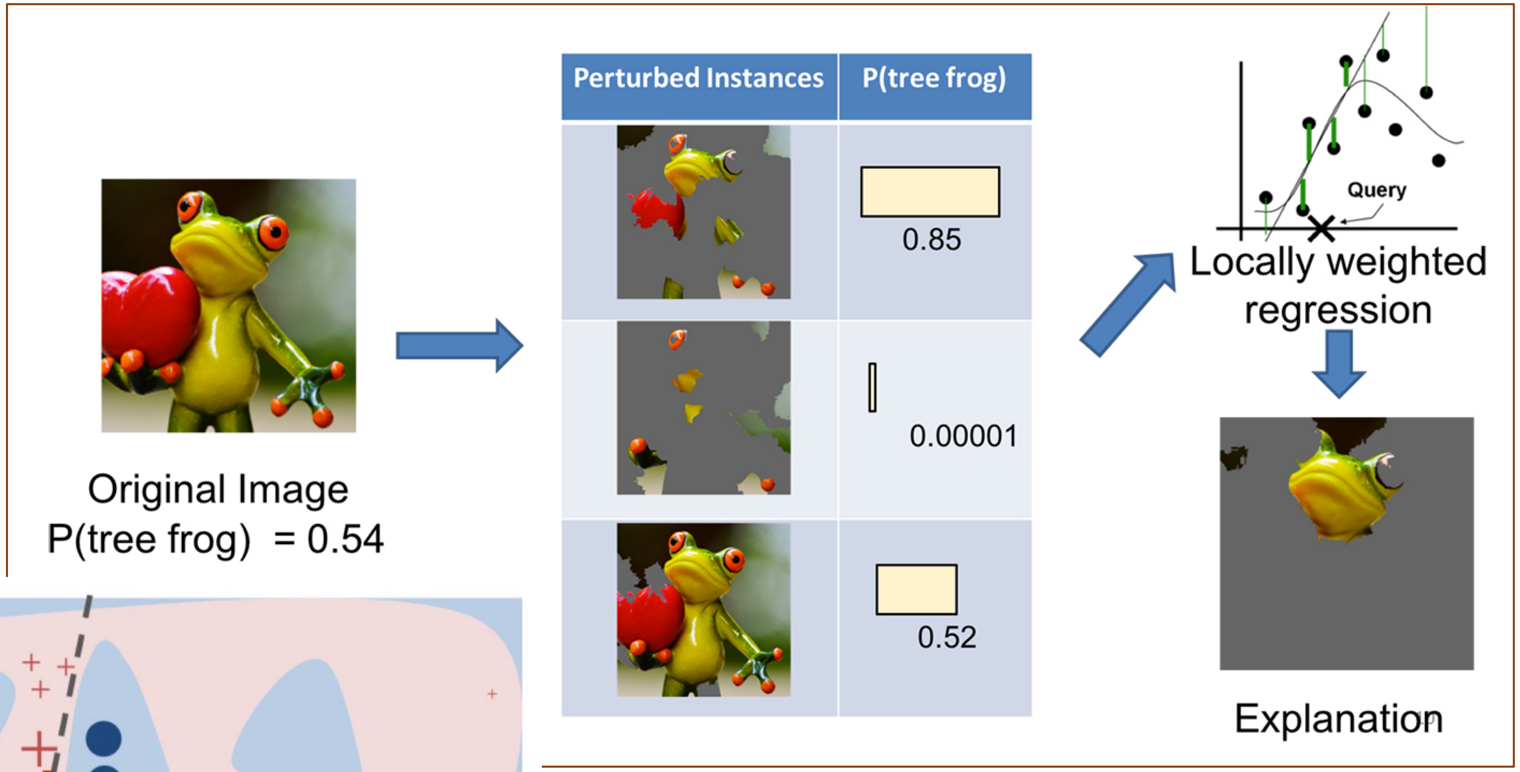
- Convolution layers
- Pooling or subsampling layers
- Fully connected layers

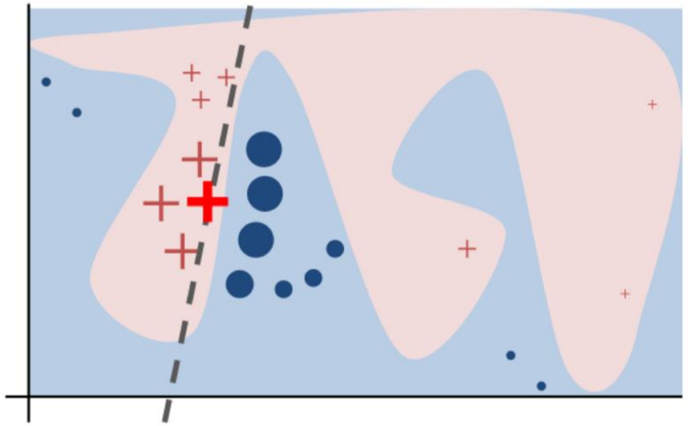


Convolutional Neural Networks



“Why Should I Trust You?” Explaining the Predictions of Any Classifier: Local Interpretable Model-agnostic Explanations (LIME) – Ribeiro et al . 2017





Local Interpretable Model-Agnostic Explanations (LIME)

Source: Ribeiro et al. 2017

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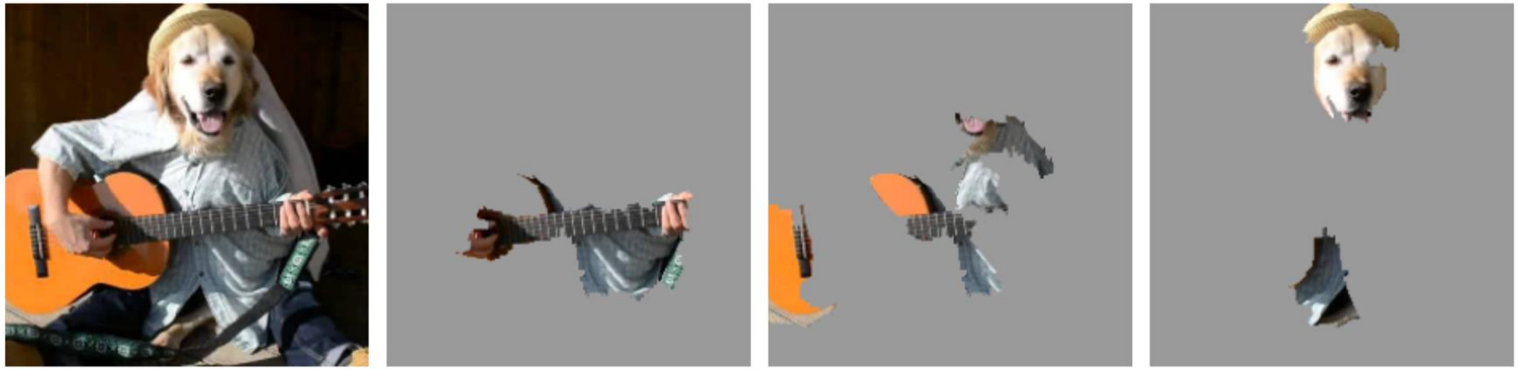


(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

Figure 4: Explaining an image classification prediction made by Google’s Inception neural network. The top 3 classes predicted are “Electric Guitar” ($p = 0.32$), “Acoustic guitar” ($p = 0.24$) and “Labrador” ($p = 0.21$)

(a) Husky classified as wolf (b) Explanation

Figure 11: Raw data and explanation of a bad model’s prediction in the “Husky vs Wolf” task.

Source: Ribeiro et al. 2017

“Why Should I Trust You?” Explaining the Predictions of Any Classifier: Local Interpretable Model-agnostic Explanations (LIME) – Ribeiro et al . 2017

Local Interpretable Model-Agnostic Explanations (LIME)

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \; \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

Proximity measure
Model being interpreted
complexity measure
How well g ~ f
Potentially interpretable model

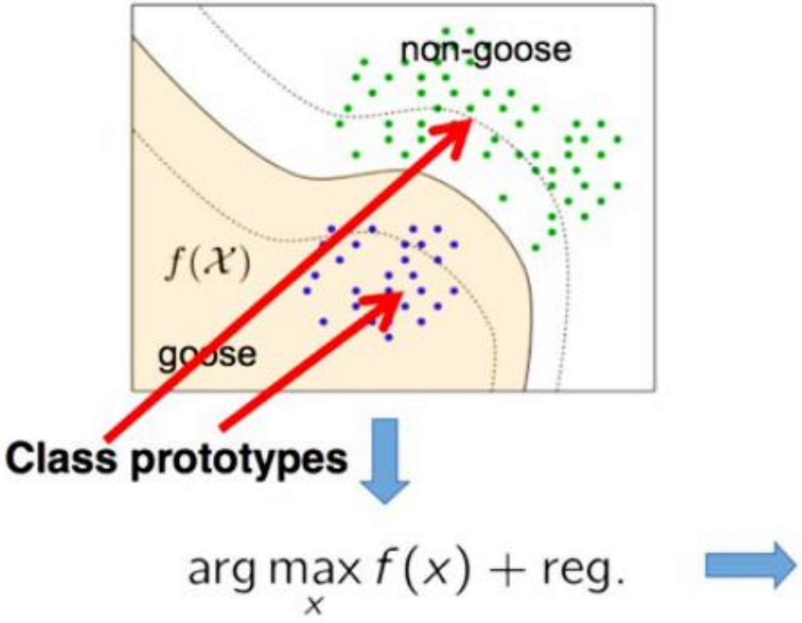
Algorithm 1 Sparse Linear Explanations using LIME

Require: Classifier f, Number of samples N
Require: Instance x, and its interpretable version x'
Require: Similarity kernel π_x, Length of explanation K
$\mathcal{Z} \leftarrow \{\}$
for $i \in \{1, 2, 3, ..., N\}$ **do**
$z'_i \leftarrow sample\_around(x')$
$\mathcal{Z} \leftarrow \mathcal{Z} \cup \langle z'_i, f(z_i), \pi_x(z_i) \rangle$
end for
$w \leftarrow$ K-Lasso$(\mathcal{Z}, K)$ $\triangleright$ with z'_i as features, $f(z)$ as target
return w

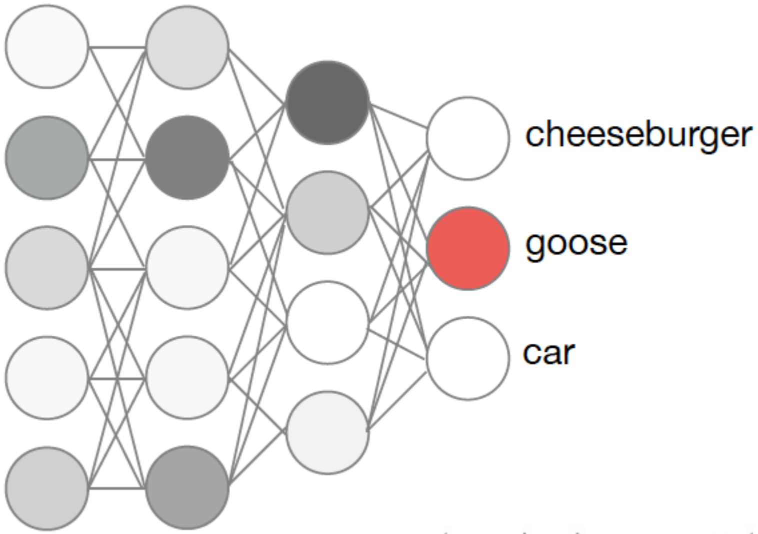
Source: Ribeiro et al. 2017

Explaining by finding prototype class members

- Global explanation technique
- Find pattern maximizing class activation
- Not-necessarily a real image

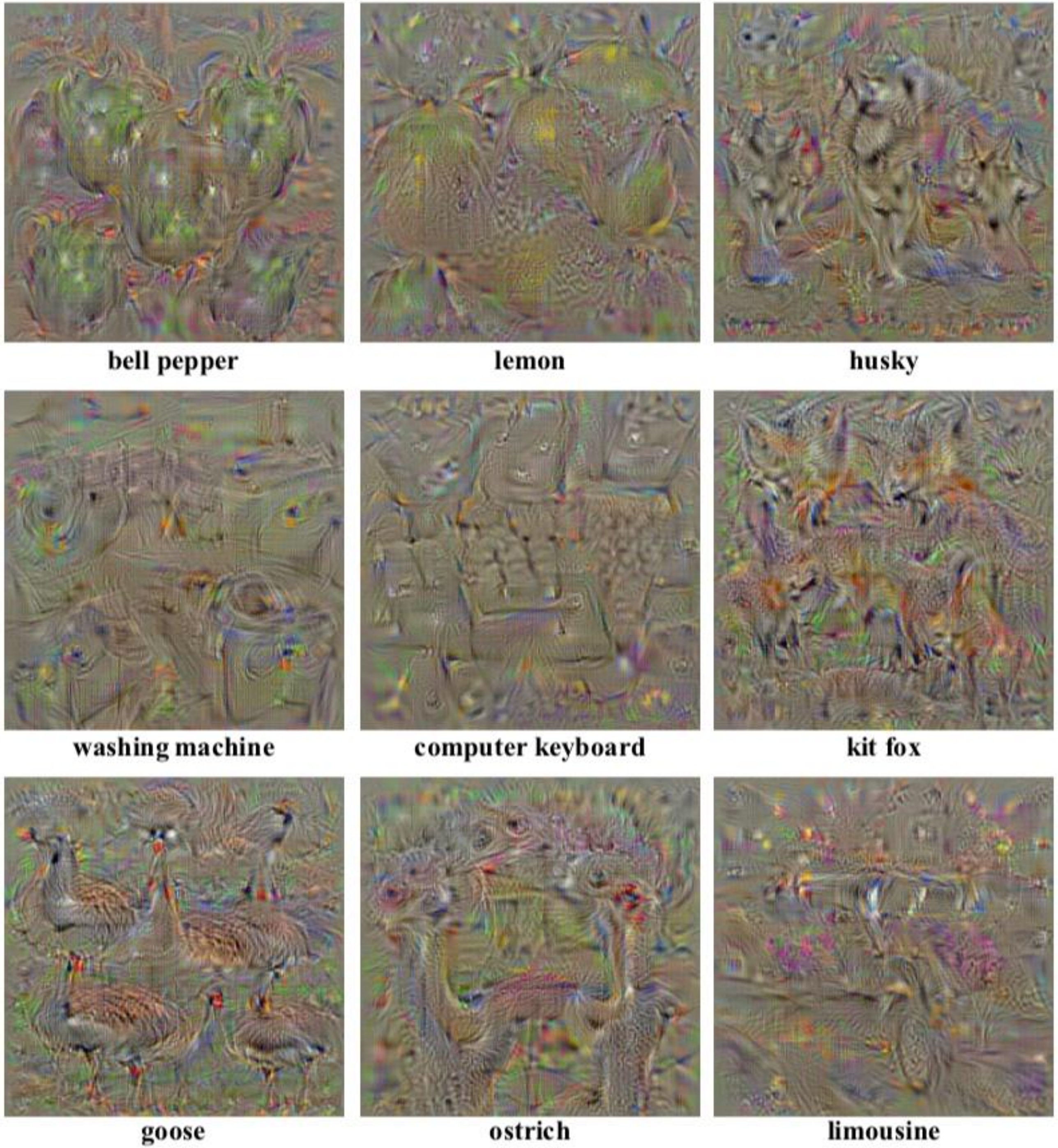


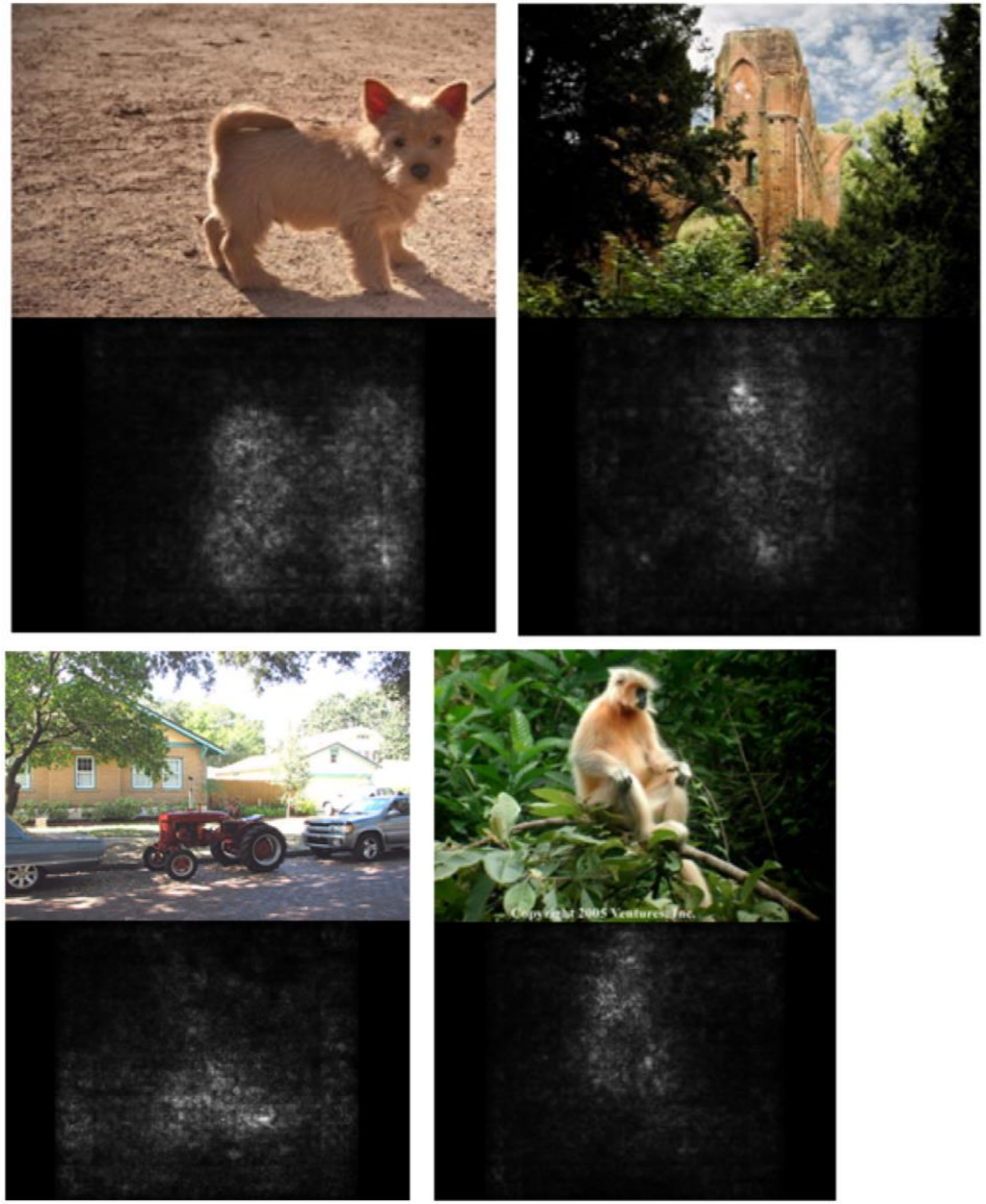
Simonyan et al. 2013



$$\max_{x \in \mathcal{X}} p_\theta(\omega_c \mid x) + \lambda \Omega(x)$$

Simonyan et al. 2013. Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps

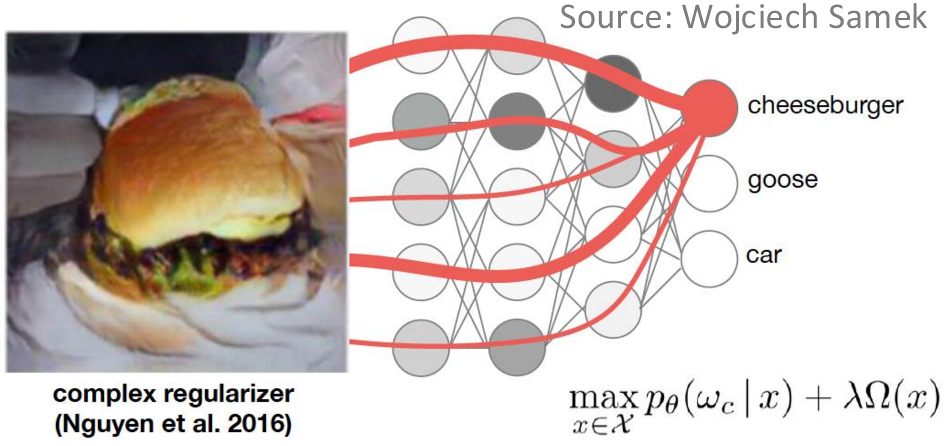




Simonyan et al. 2013. Deep Inside Convolutional Networks:
Visualising Image Classification Models and Saliency Maps

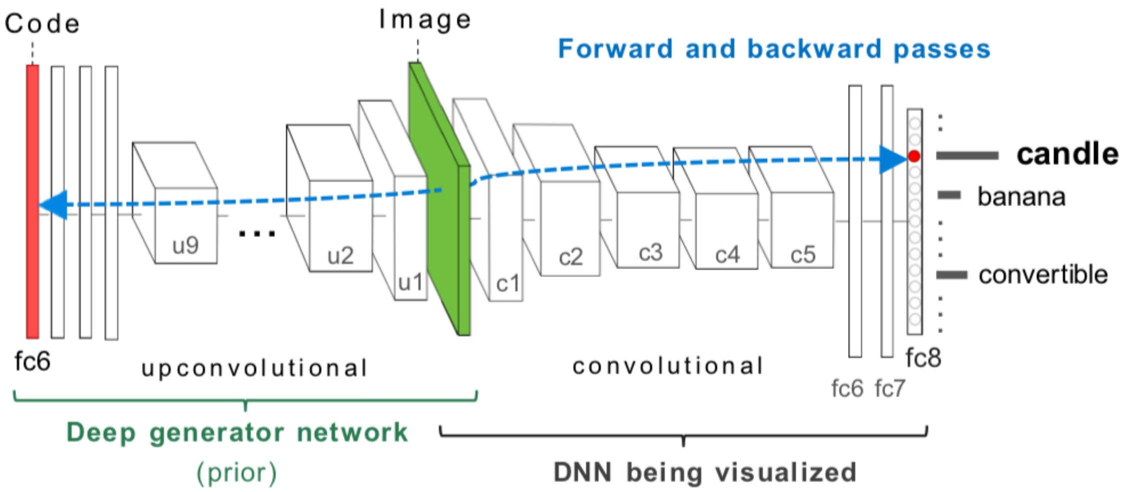
Synthesizing the preferred inputs for neurons in neural networks via deep generator networks
- Nguyen et al. 2016

Source: Wojciech Samek



Regularization via generative adversarials

1.- Forwards pass of gradients
2- Detect maximum activation
3.- Backwark pass of “FC6” into encoder



Feature vector

classifier

generator

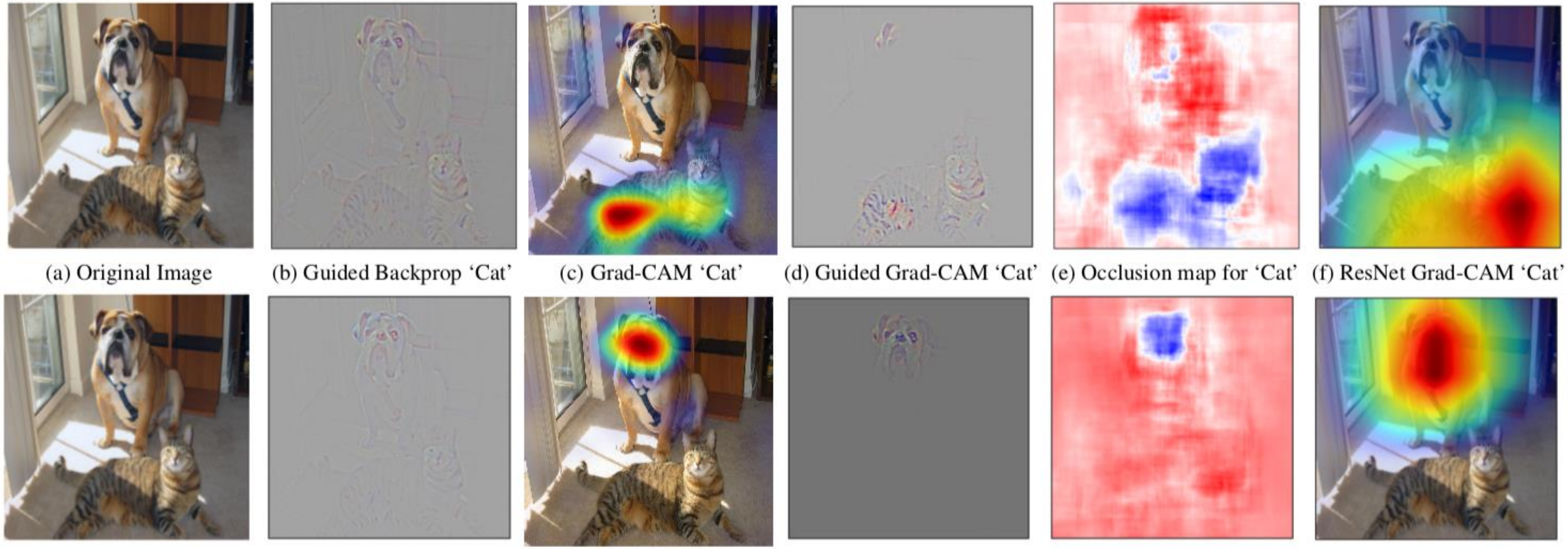
$$\widehat{\mathbf{y}^l} = \arg\max_{\mathbf{y}^l}(\Phi_h(G_l(\mathbf{y}^l)) - \lambda\|\mathbf{y}^l\|)$$

- Prior generator and DNN can be trained separately

Synthesizing the preferred inputs for neurons in neural networks via deep generator networks
 - Nguyen et al. 2016

Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization - Selvaraju et al. 2017

- Generalization of CAM: Zhou et al. Learning Deep Features for Discriminative Localization. In *CVPR*, 2016.
  - Applicable to any CNN-based (CAM requires conv feature maps → global average pooling → softmax layer)
- Motivation – Good visualization is class-specific <u>and</u> detailed

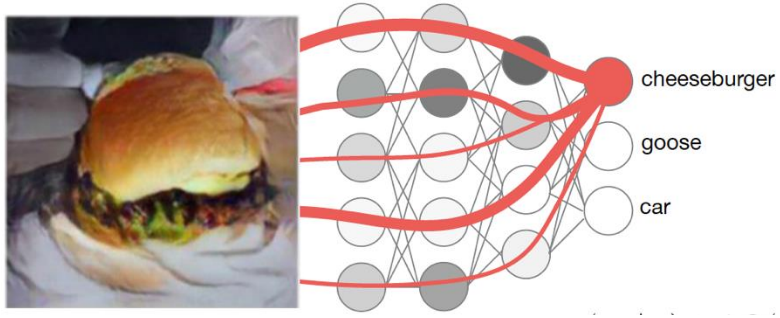


(a) Original Image (b) Guided Backprop 'Cat' (c) Grad-CAM 'Cat' (d) Guided Grad-CAM 'Cat' (e) Occlusion map for 'Cat' (f) ResNet Grad-CAM 'Cat'

(g) Original Image (h) Guided Backprop 'Dog' (i) Grad-CAM 'Dog' (j) Guided Grad-CAM 'Dog' (k) Occlusion map for 'Dog' (l) ResNet Grad-CAM 'Dog'

Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization - Selvaraju et al. 2017

Single forward and (partial) backward pass



Source: Wojciech Samek

Neuron importance weights $\alpha_k^c = \overbrace{\frac{1}{Z}\sum_i \sum_j}^{\text{global average pooling}} \underbrace{\frac{\partial y^c}{\partial A_{ij}^k}}_{\text{gradients via backprop}}$

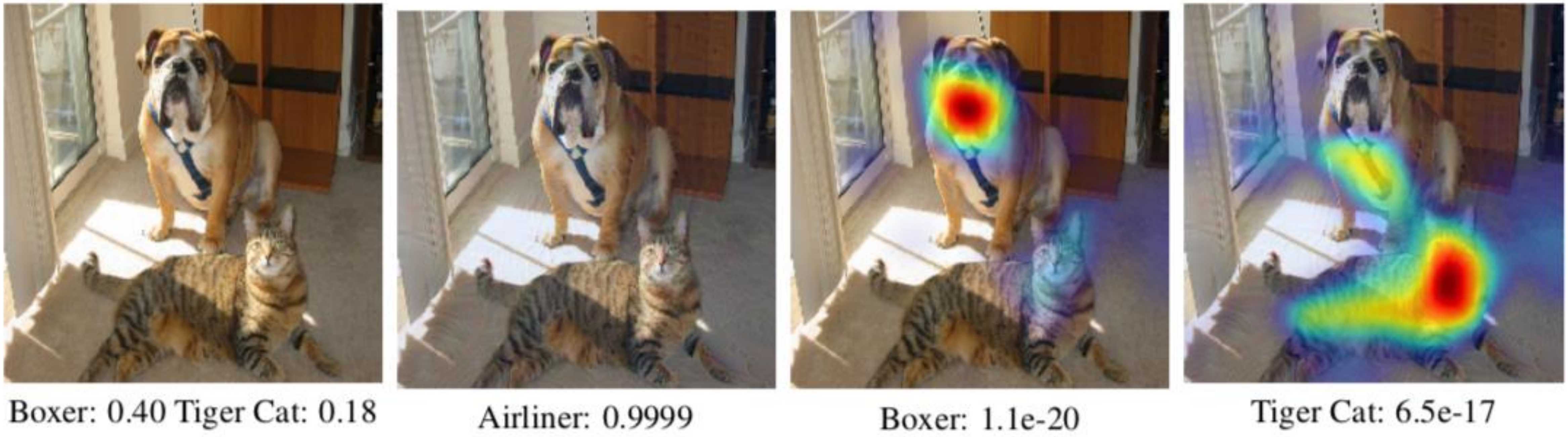
A^k: kth feature map

y^c: score for class c (before softmax)

$$L_{\text{Grad-CAM}}^c = ReLU \underbrace{\left(\sum_k \alpha_k^c A^k\right)}_{\text{linear combination}}$$

Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization - Selvaraju et al. 2017

Robustness to <u>adversarial examples</u>: Examples where VGG got "fooled"



(a) Original image (b) Adversarial image (c) Grad-CAM "Dog" (d) Grad-CAM "Cat"

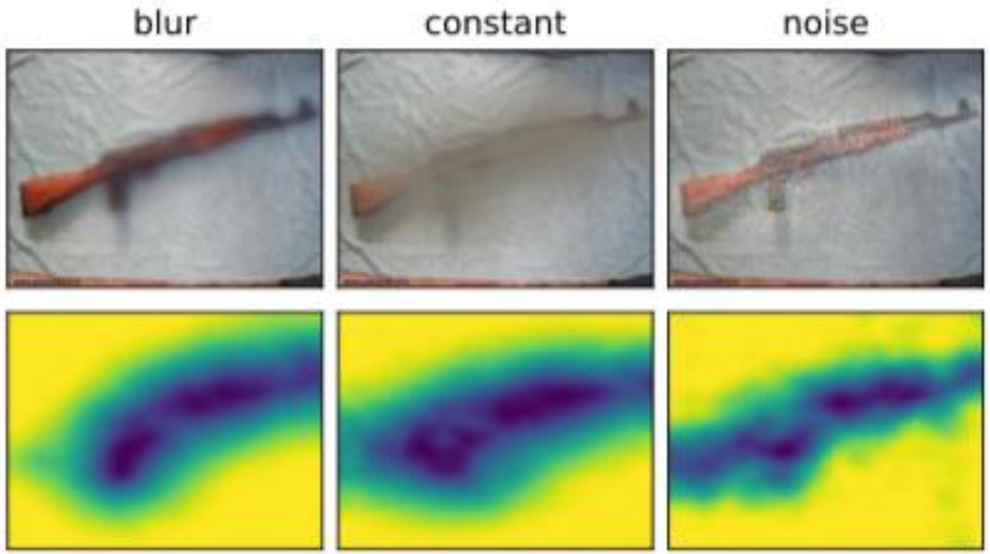
Interpretable Explanations of Black Boxes by Meaningful Perturbation - Fong et al. 2018

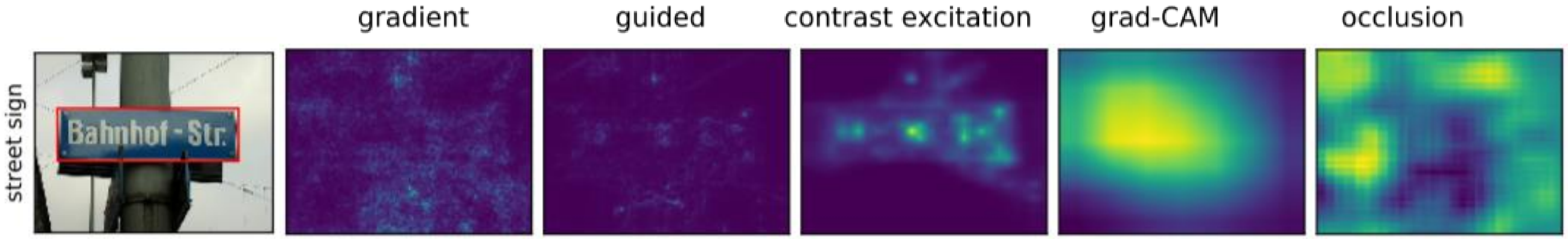
Motivation: Saliency (gradient-based) approaches are not specific enough

Key idea: select the right perturbation* to study the effect on the prediction function f(x)

Meaning of an explanation depends on the meaning of the changes applied to the input x

*Perturbation ←→ type of information deletion





Interpretable Explanations of Black Boxes by Meaningful Perturbation - Fong et al. 2018

Optimal ”information deletion” mask

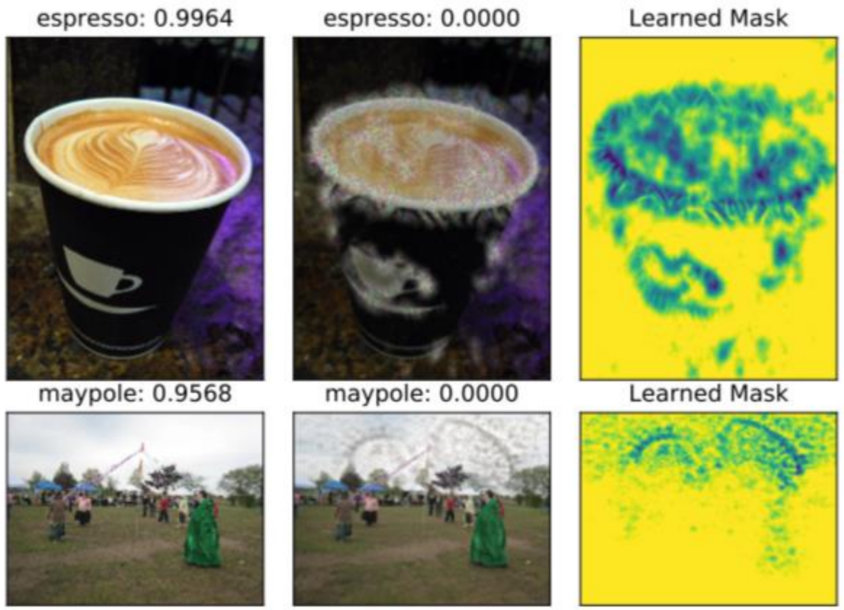
Enforces minimal mask size

perturbator

Input image

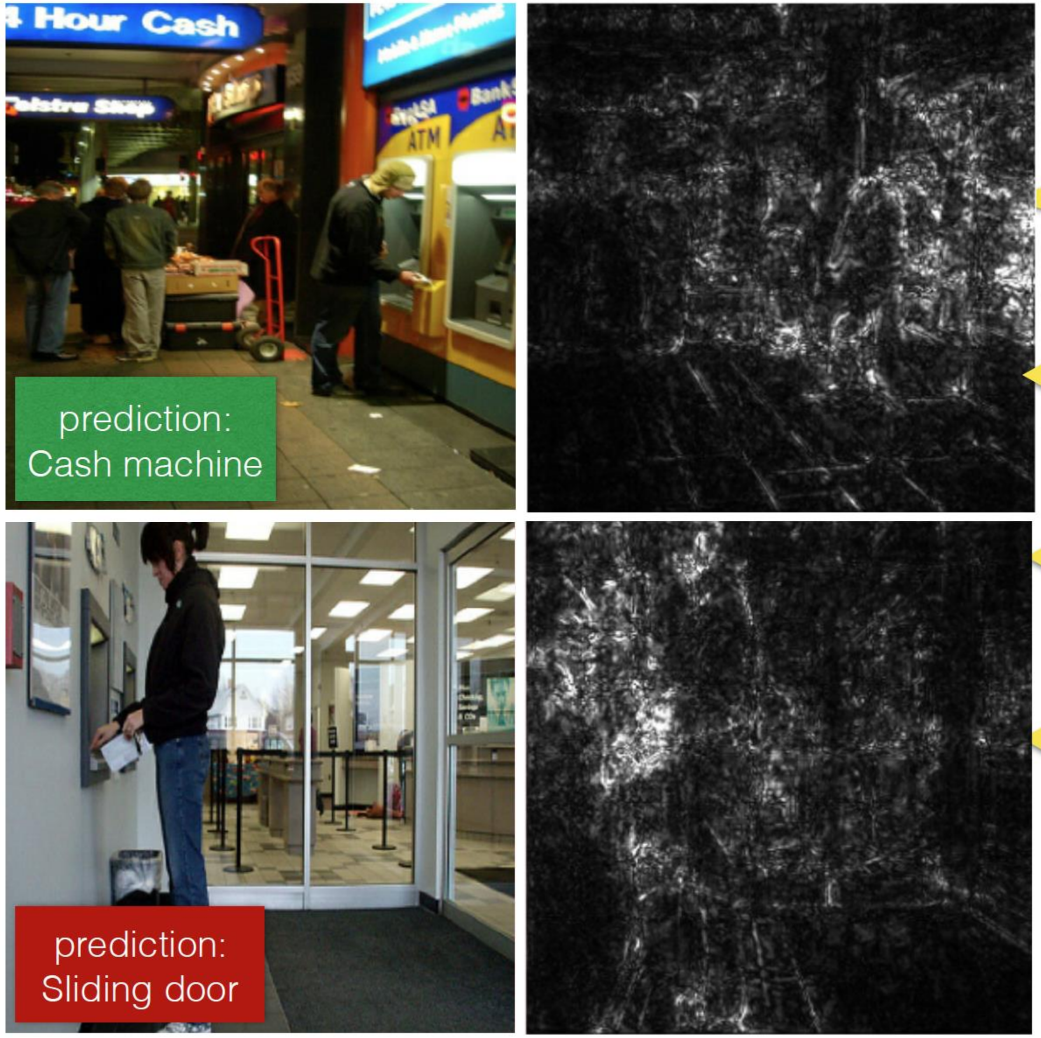
$$m^* = \underset{m\in[0,1]^\Lambda}{\operatorname{argmin}} \lambda\|\mathbf{1} - m\|_1 + f_c(\Phi(x_0; m))$$

Predictor for class c being interpreted



Can lead to unnatural perturbations → against explanations

Testing with Concept Activation Vectors – TCAV – Kim et al. 2017



Were there more pixels on the cash machine than on the person?

Did the 'human' concept matter?
Did the 'glasses' or 'paper' matter?

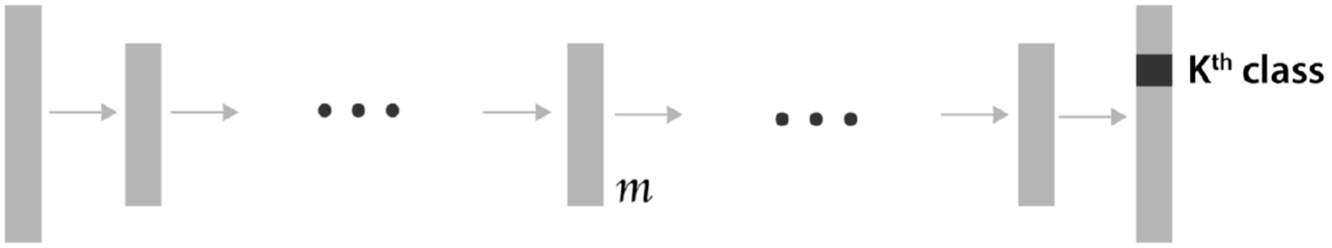
Which concept mattered more?

Is this true for all other cash machine predictions?

Wouldn't it be great if we can **quantitatively** measure how important *any* of these **user-chosen concepts** are?

Image source: Kim Been

Testing with Concept Activation Vectors – TCAV – Kim et al. 2017



Quantitative explanation: how much a concept (e.g., gender, race) was important for a prediction in a trained model.

…even if the concept was not part of the training.

Image source: Kim Been

Testing with Concept Activation Vectors – TCAV – Kim et al. 2017

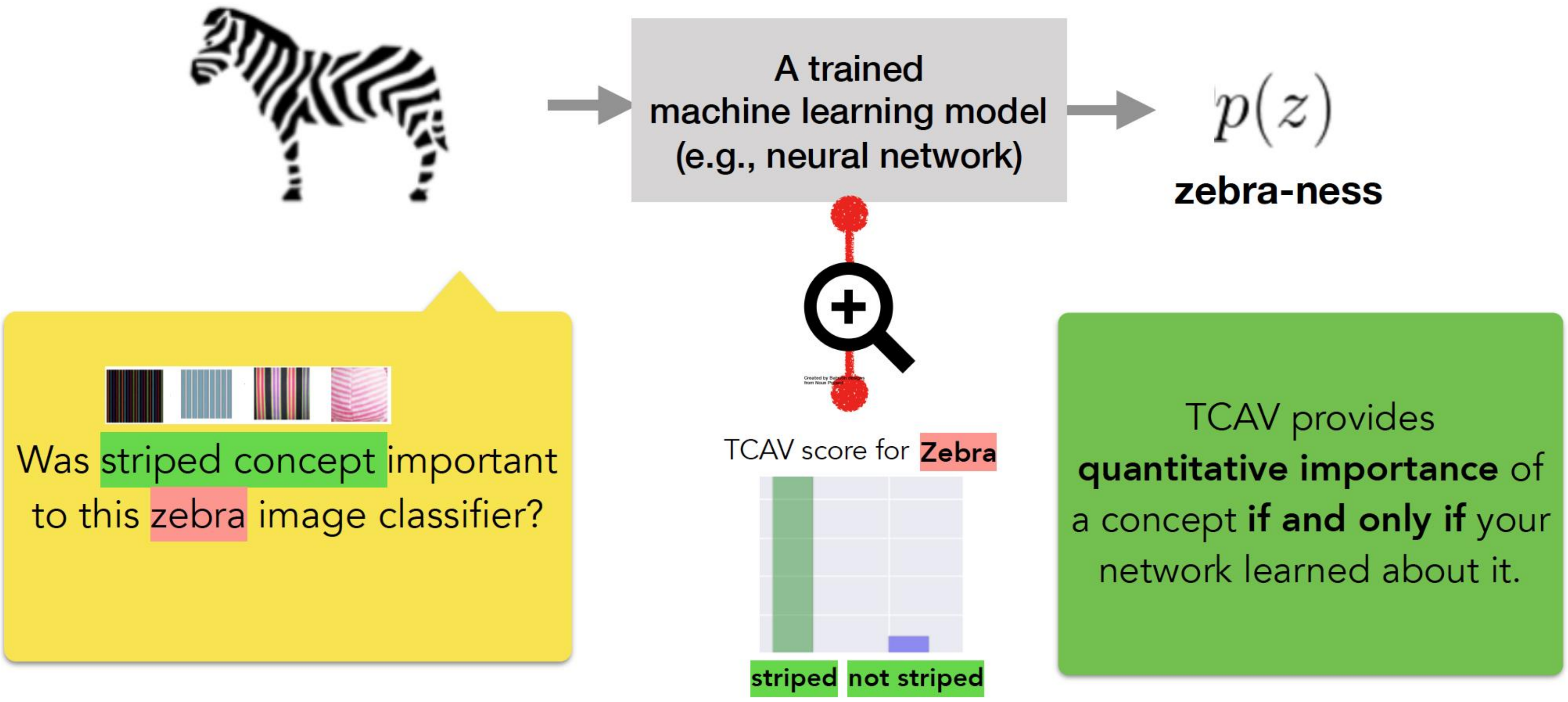


Image source: Kim Been

Testing with Concept Activation Vectors – TCAV – Kim et al. 2017

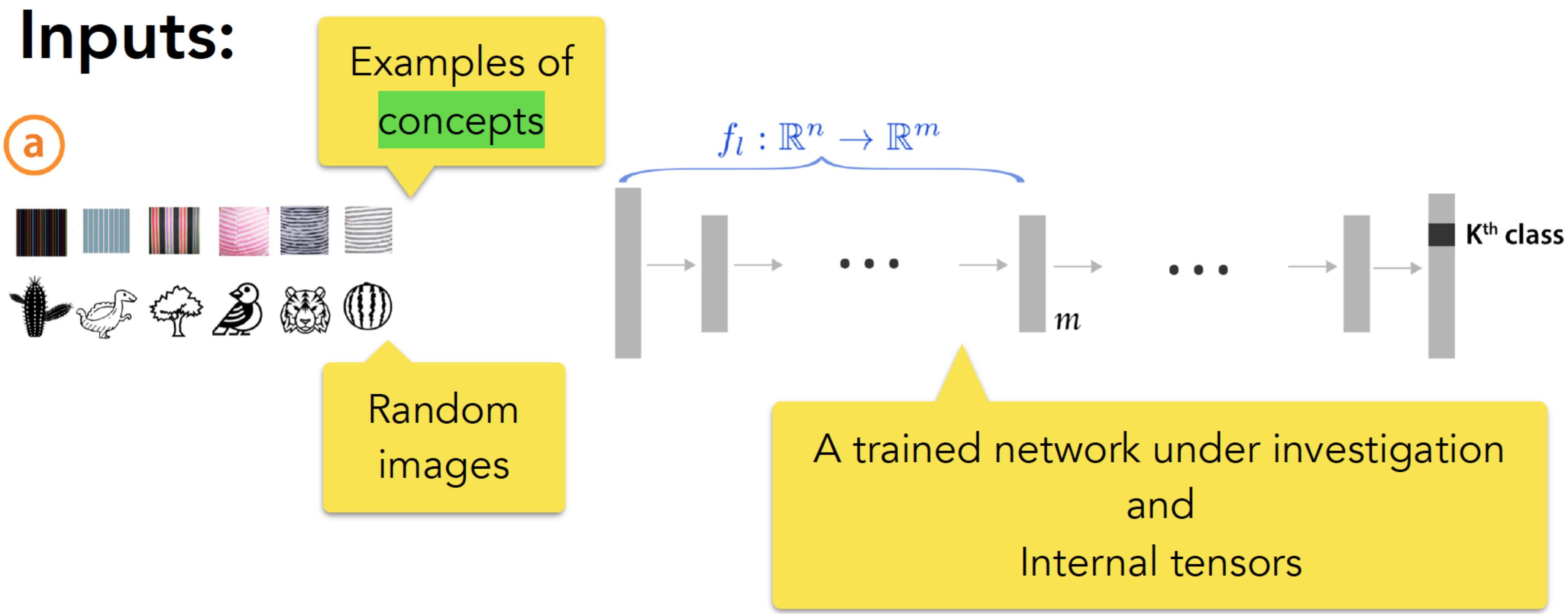
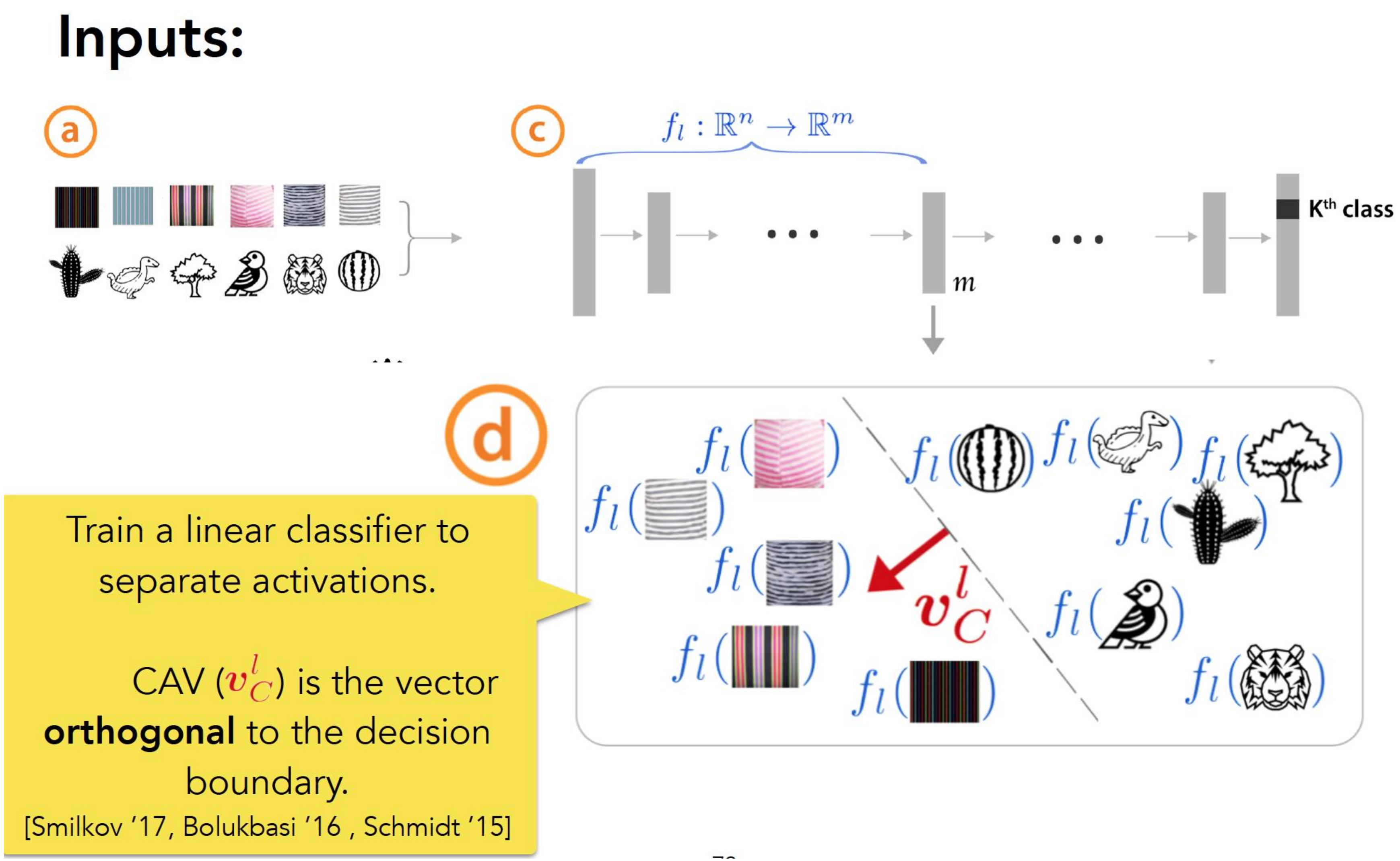


Image source: Kim Been

Testing with Concept Activation Vectors – TCAV – Kim et al. 2017



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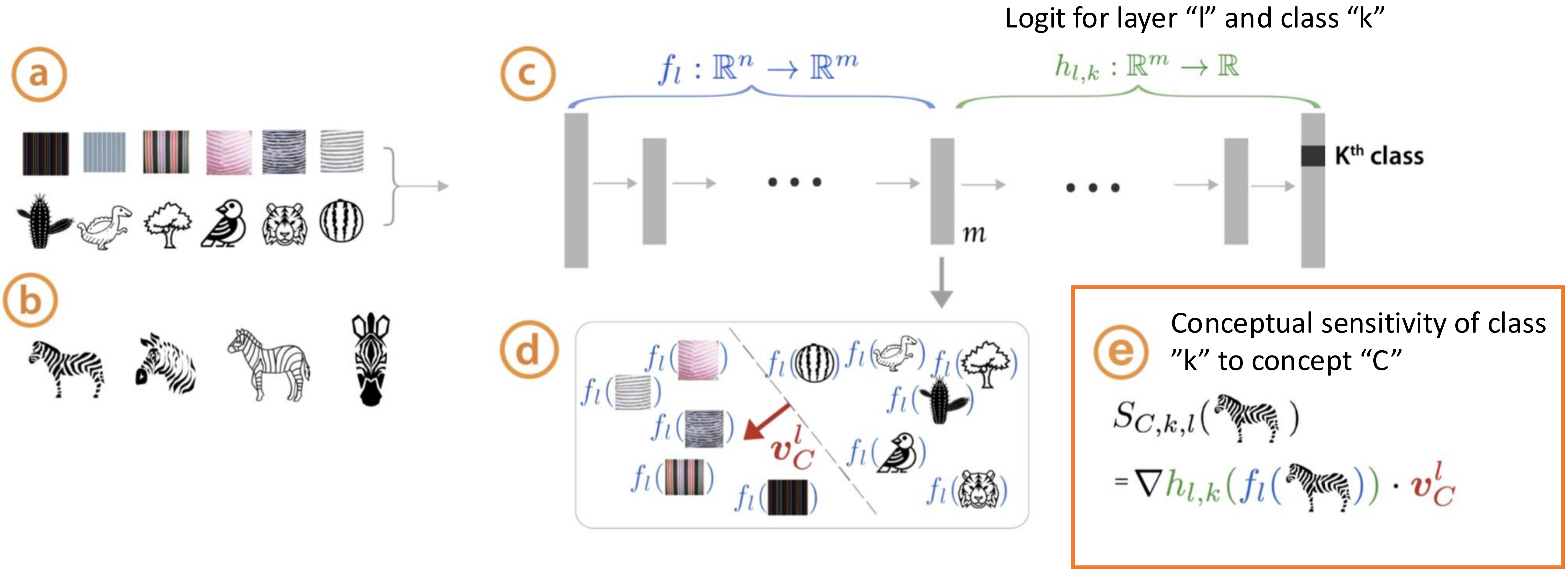
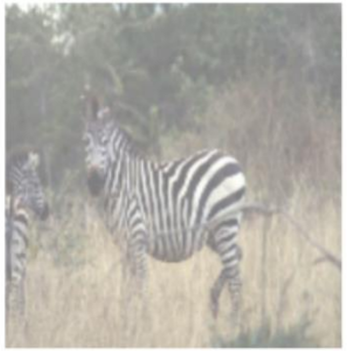
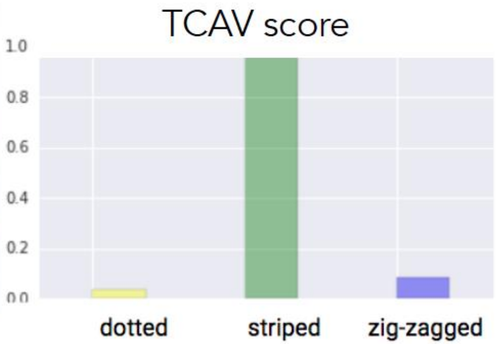


Image source: Kim Been

Testing with Concept Activation Vectors – TCAV – Kim et al. 2017

TCAV





zebra-ness → $\partial p(z)$

striped CAV → $\partial \boldsymbol{v}_C^l$

$$\frac{\partial p(z)}{\partial \boldsymbol{v}_C^l} = S_{C,k,l}(\boldsymbol{x})$$

Directional derivative with CAV

$S_{C,k,l}($  $)$

$S_{C,k,l}($  $)$

$S_{C,k,l}($  $)$

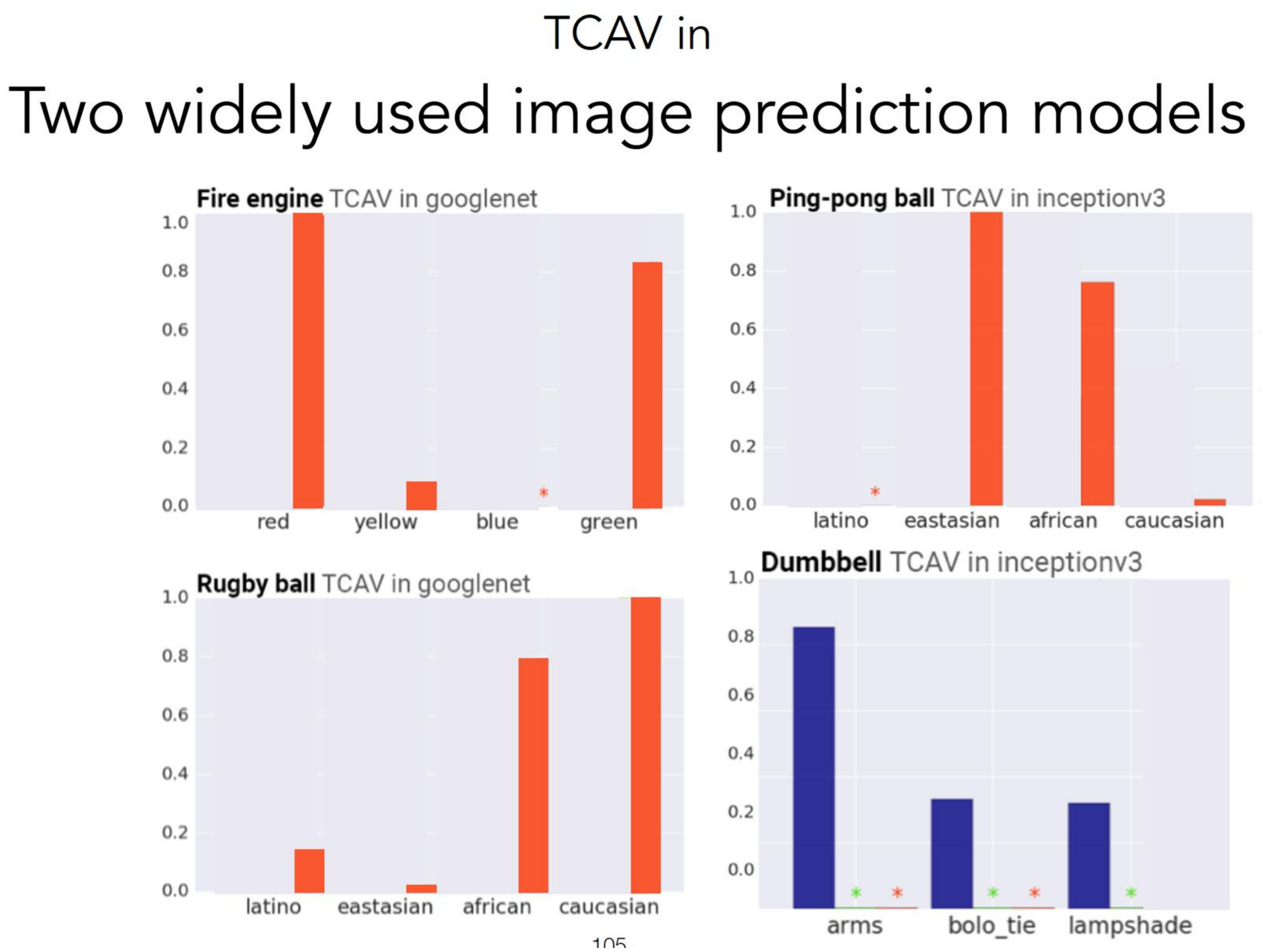
$S_{C,k,l}($  $)$

$$\boxed{\mathrm{TCAV}_{\mathrm{Q}_{C,k,l}} = \frac{|\{\boldsymbol{x} \in X_k : S_{C,k,l}(\boldsymbol{x}) > 0\}|}{|X_k|}}$$

fraction of k-class inputs whose l-layer activation vector was positively influenced by concept C

Image source: Kim Been

Testing with Concept Activation Vectors – TCAV – Kim et al. 2017



TCAV facilitates spotting of biases in the datasets

Image source: Kim Been

What is AI learning? What if this differs from human reasoning/knowledge?

(a) Husky classified as wolf (b) Explanation

Shortcut!

VS.

Artificial intelligence predicts patients' race from their medical images

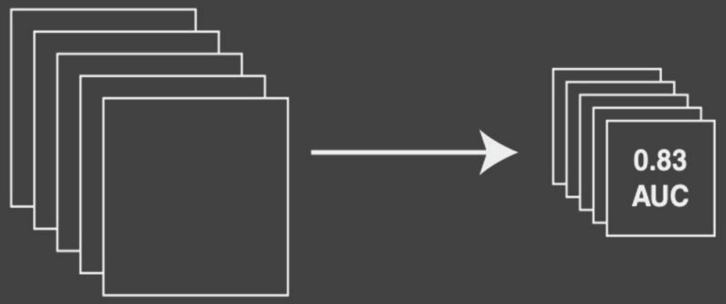
Study shows AI can identify self-reported race from medical images that contain no indications of race detectable by human experts.

Rachel Gordon | MIT CSAIL
May 20, 2022

Biases!

Knowledge discovery!

Right for the Wrong Reasons: Shortcut Learning as a Barrier to Clinical Deployment



Baniecki et al. 2024

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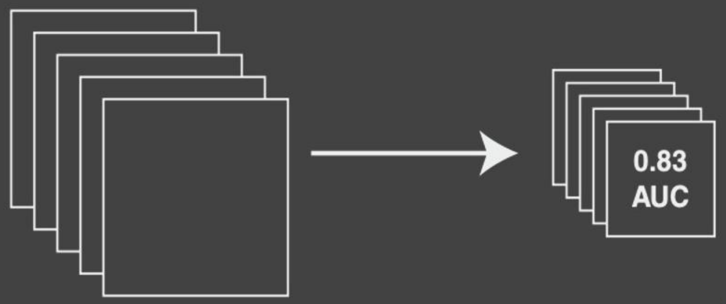

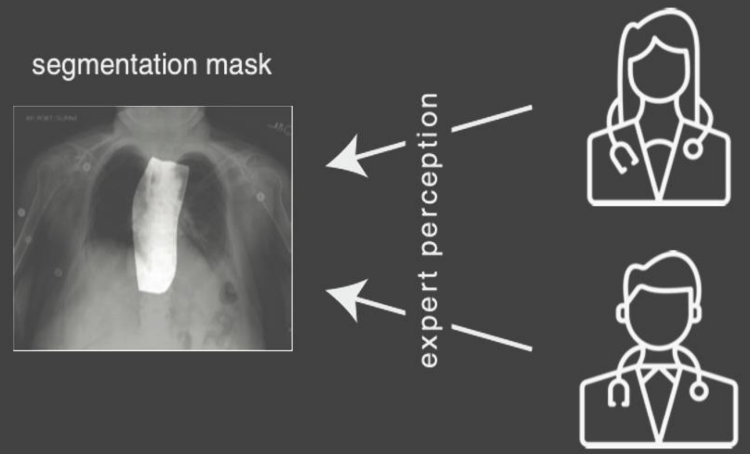

Baniecki et al. 2024

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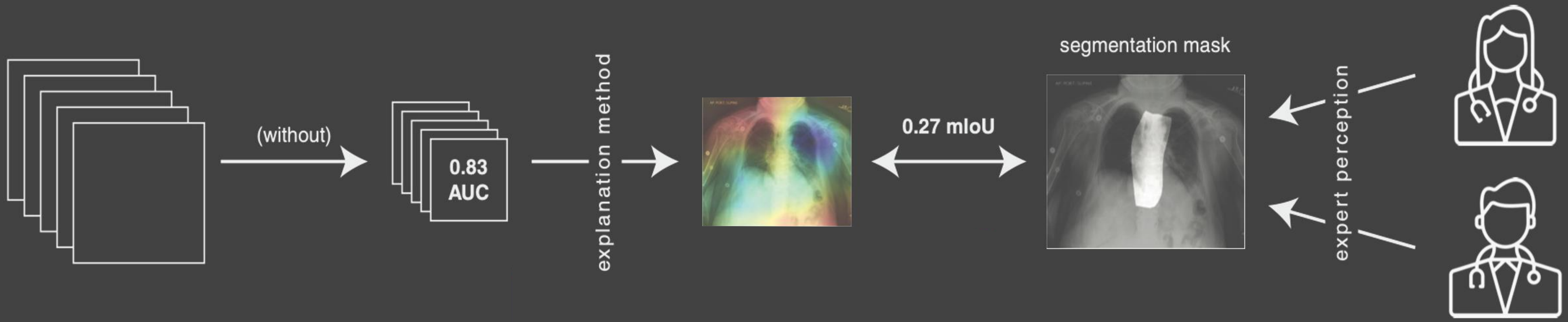
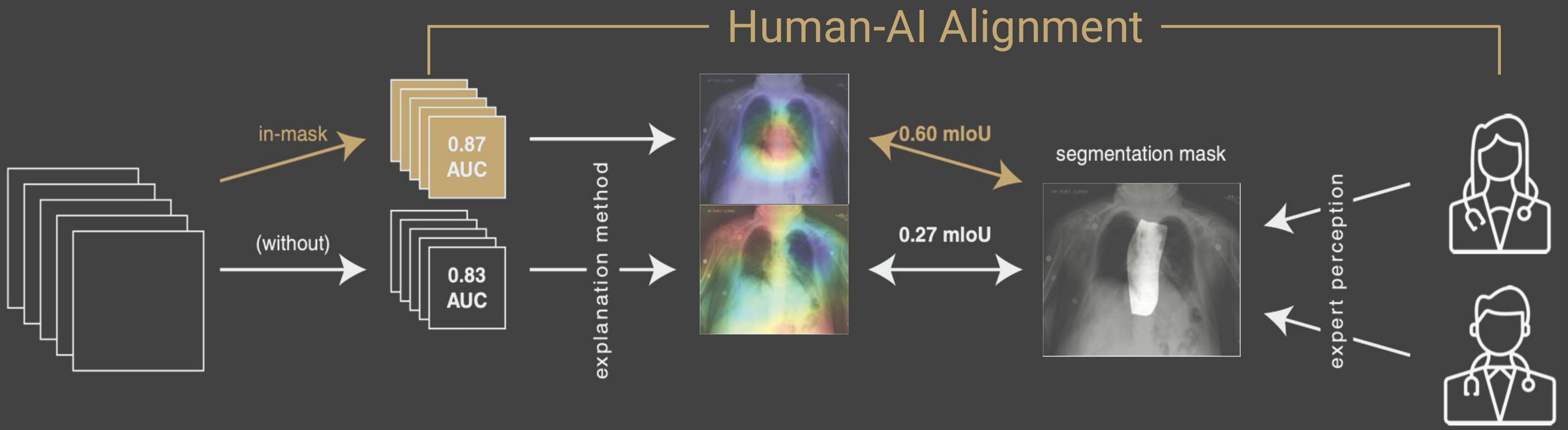


Baniecki et al. 2024

Right for the Wrong Reasons: Shortcut Learning as a Barrier to Clinical Deployment



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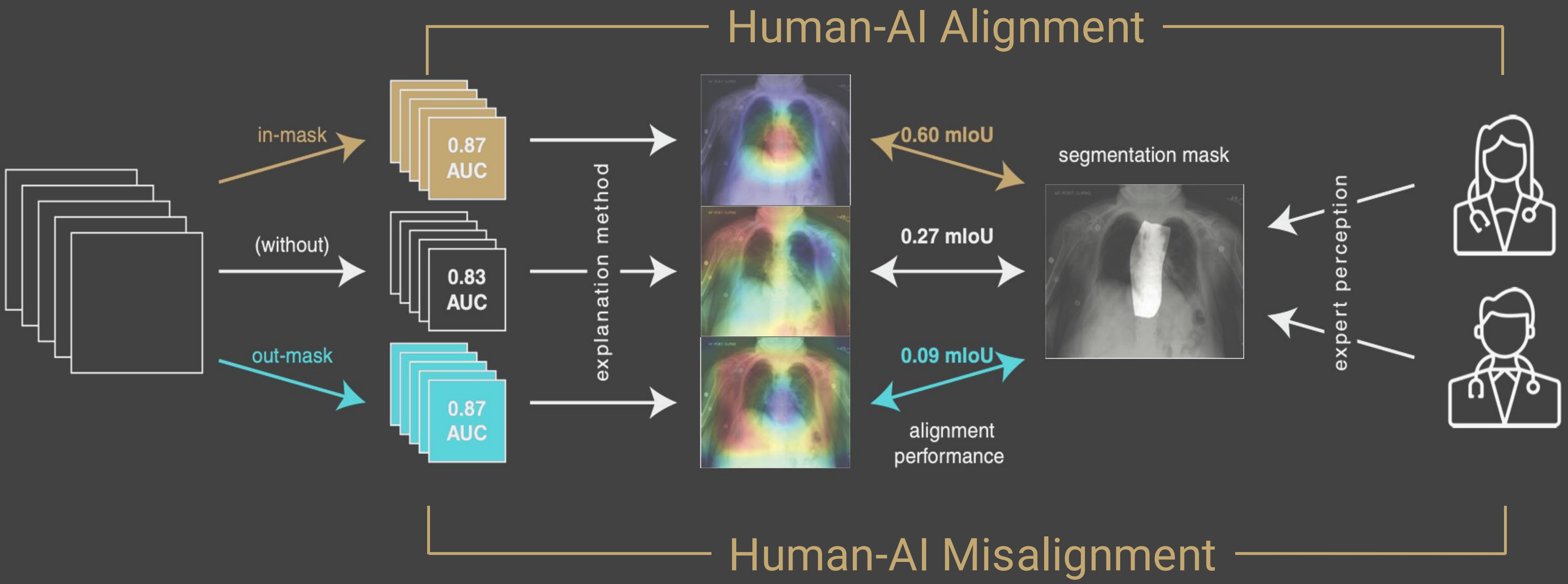
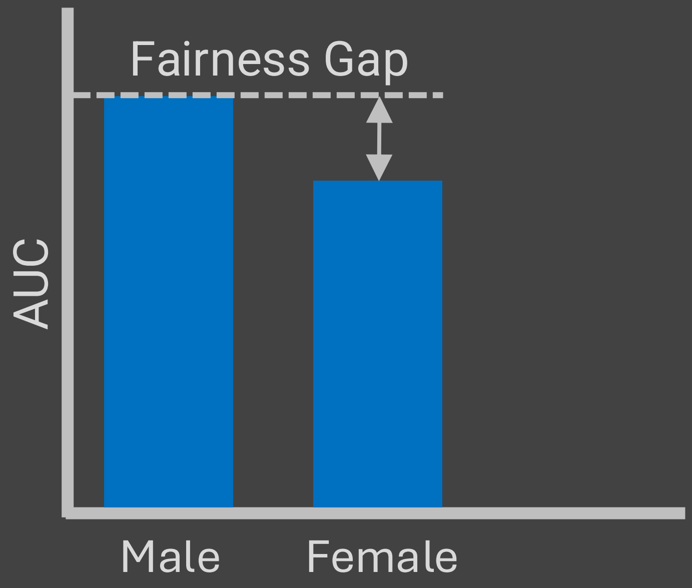


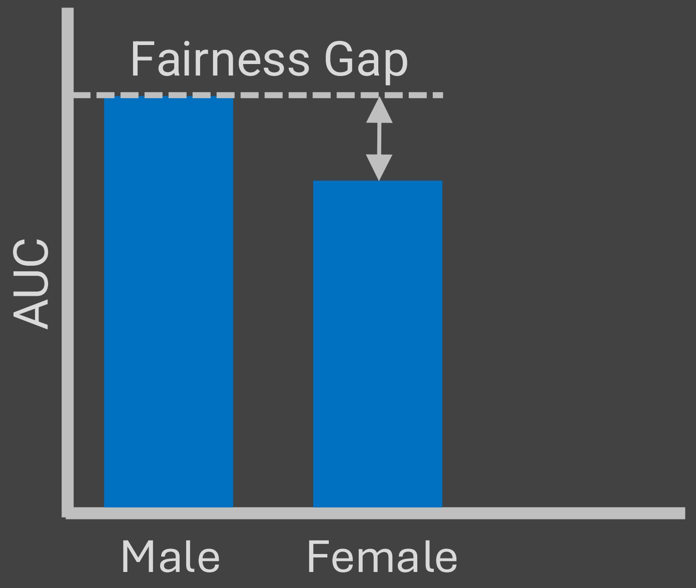
Baniecki et al. 2024

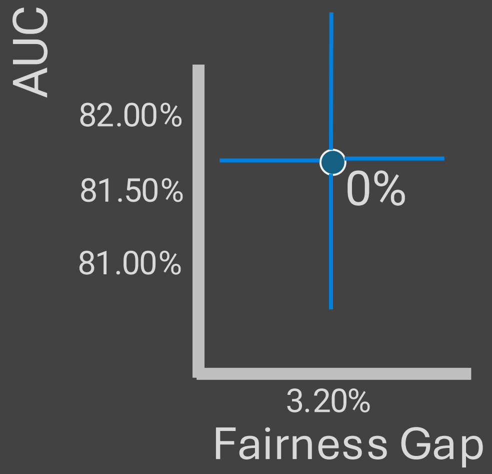
Reliable Clinical AI: Fair and Trustworthy

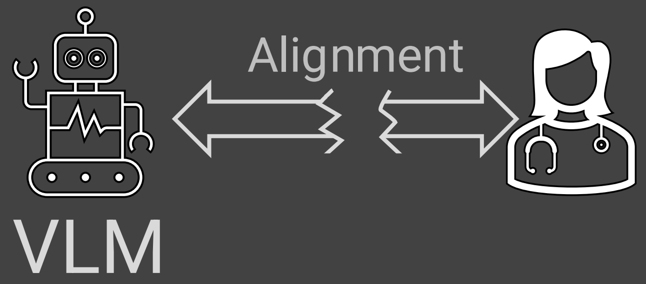


“Primum non nocere”
(First do no harm)

Reliable Clinical AI: Fair and Trustworthy







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Luo et al. Miccai’25

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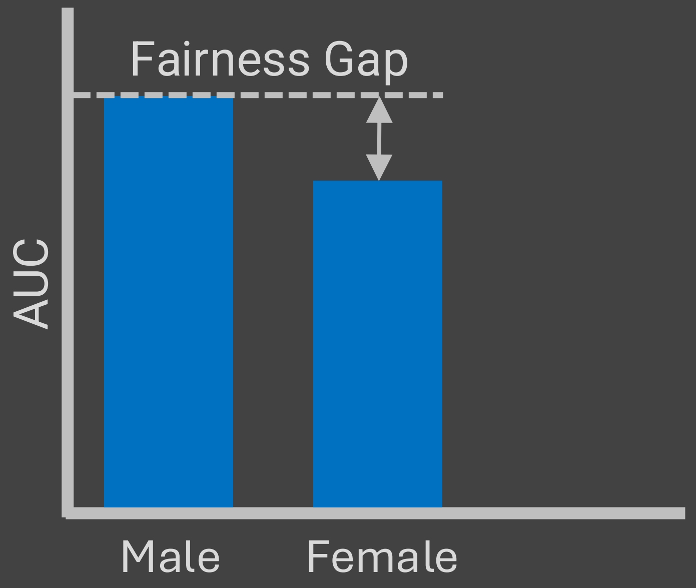

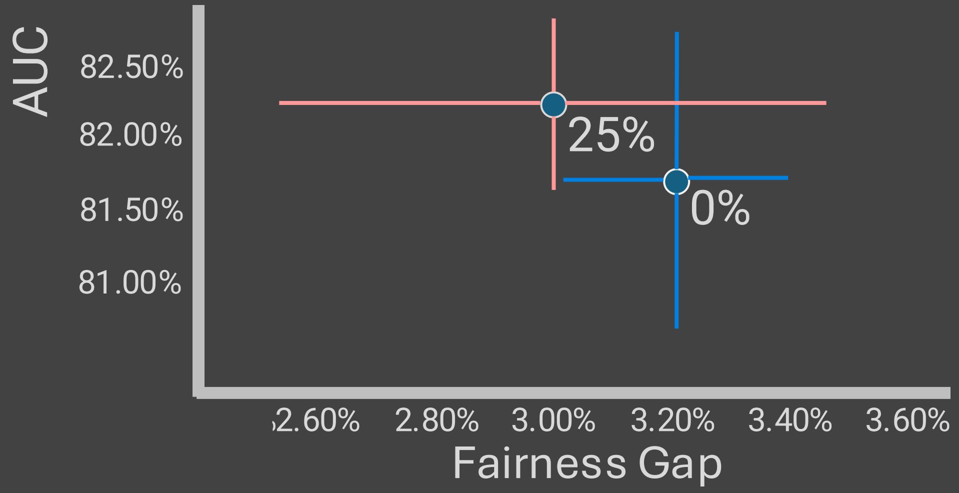

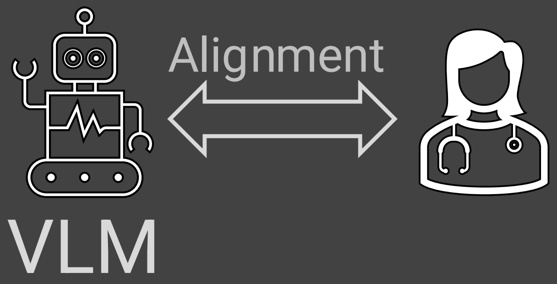

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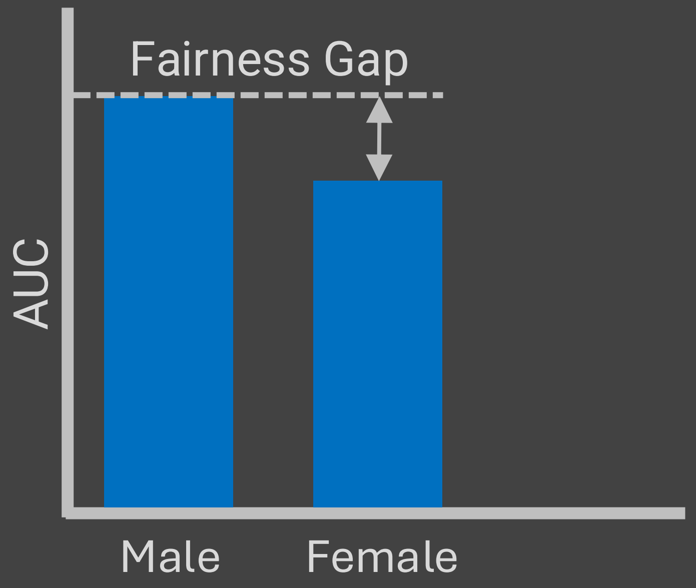

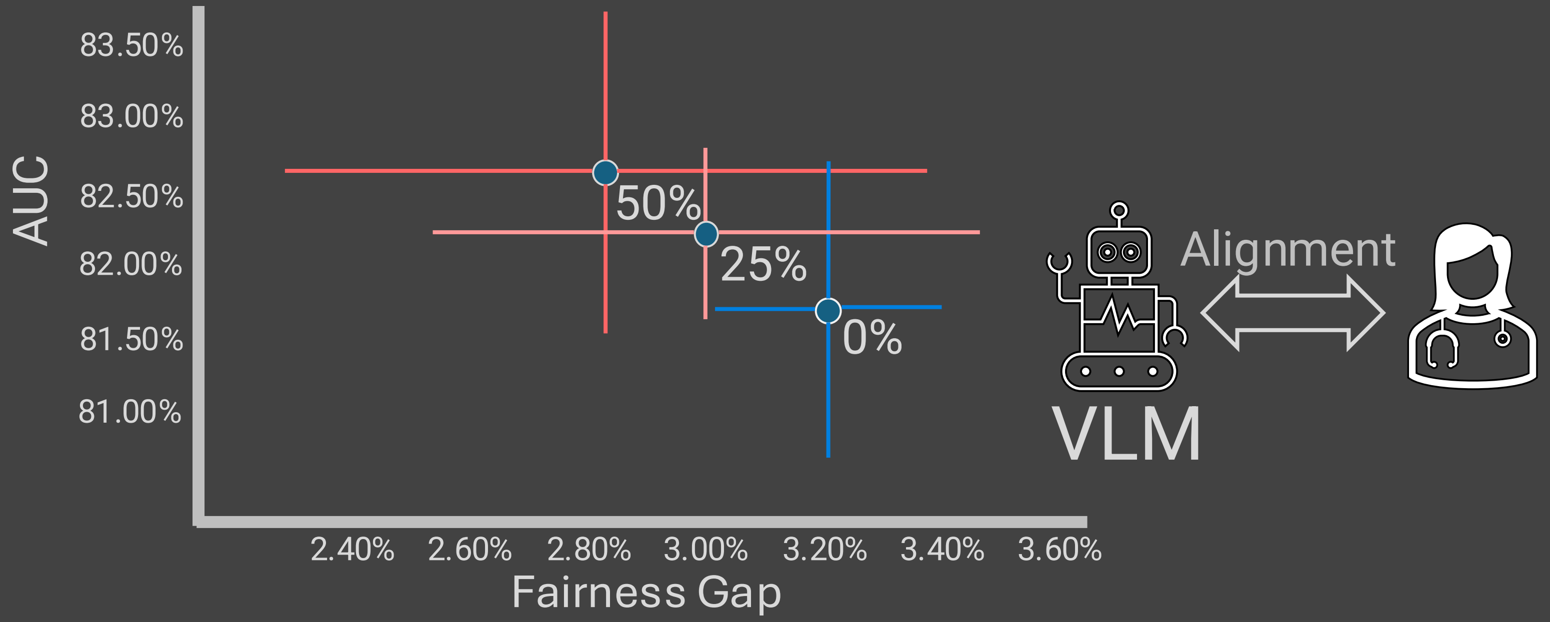

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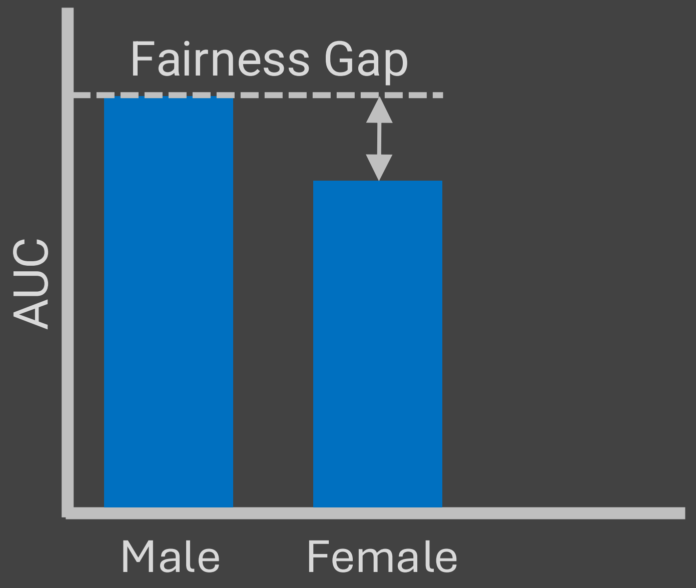

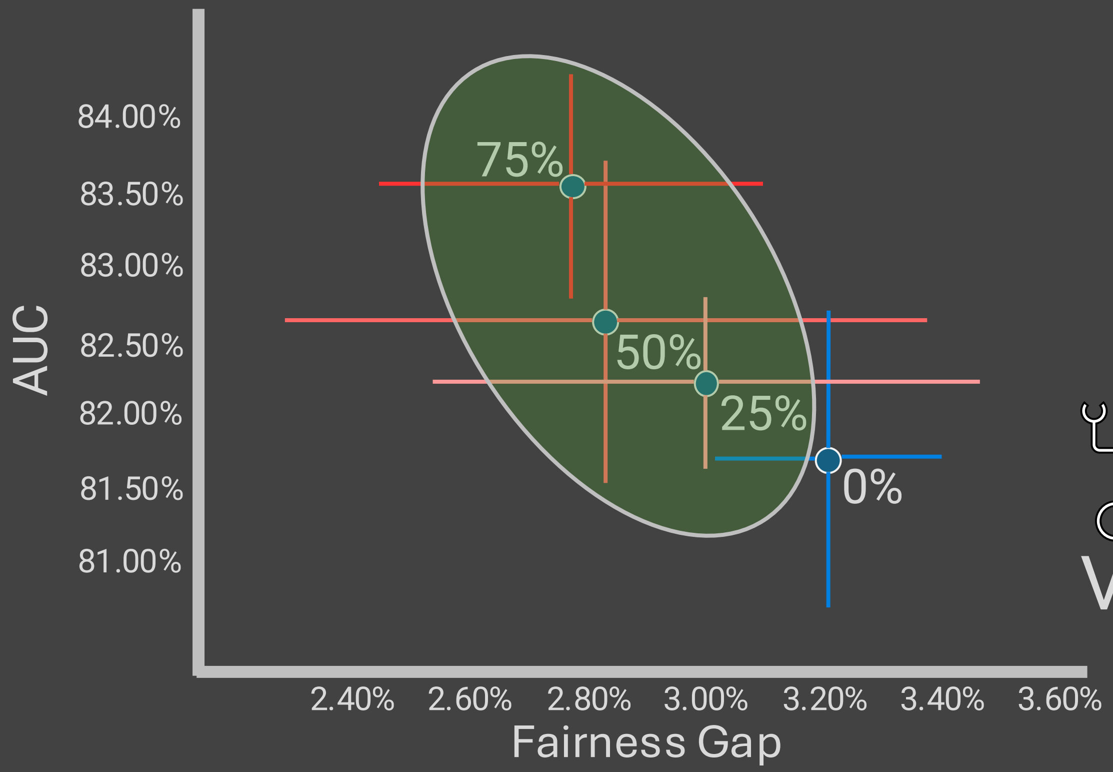

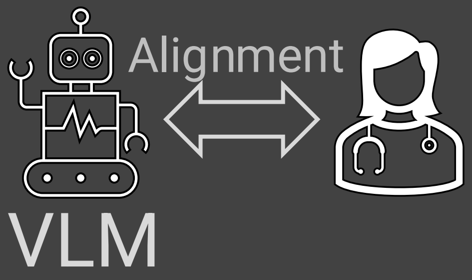

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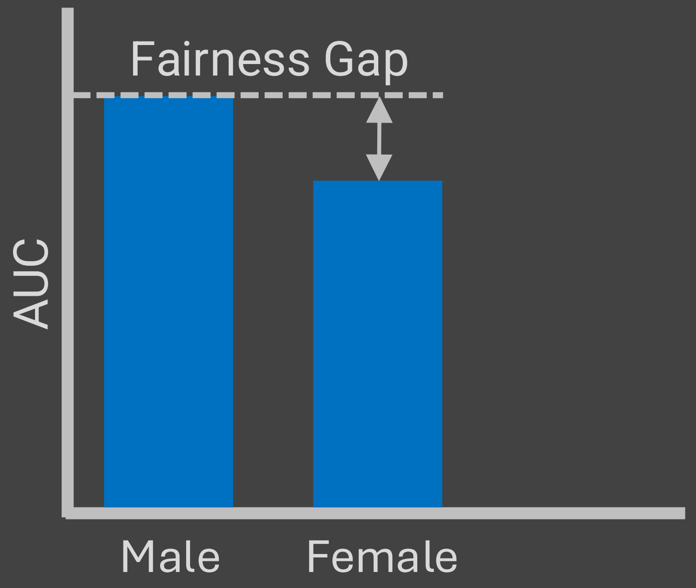

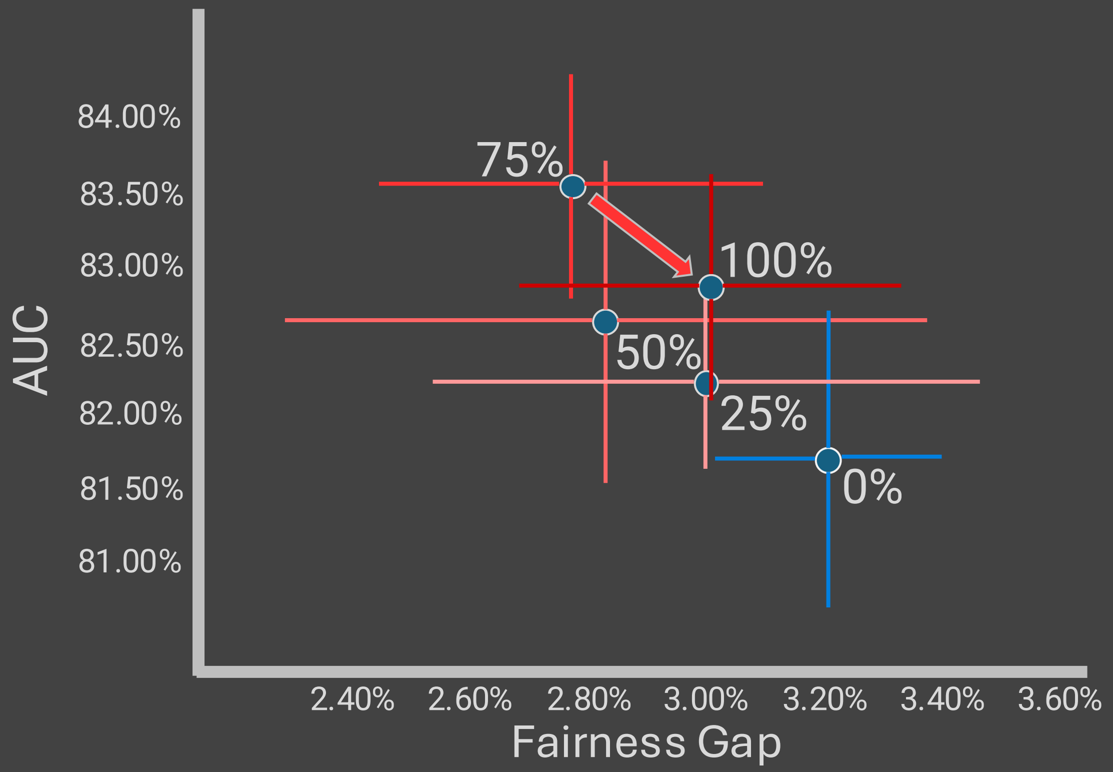

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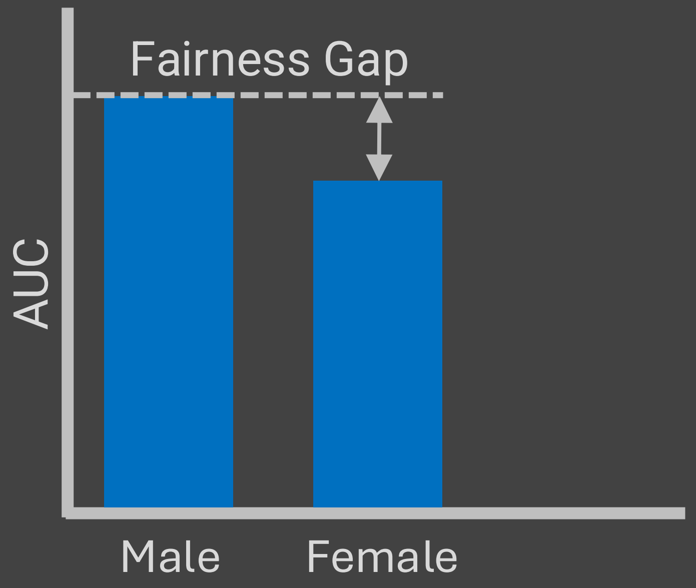

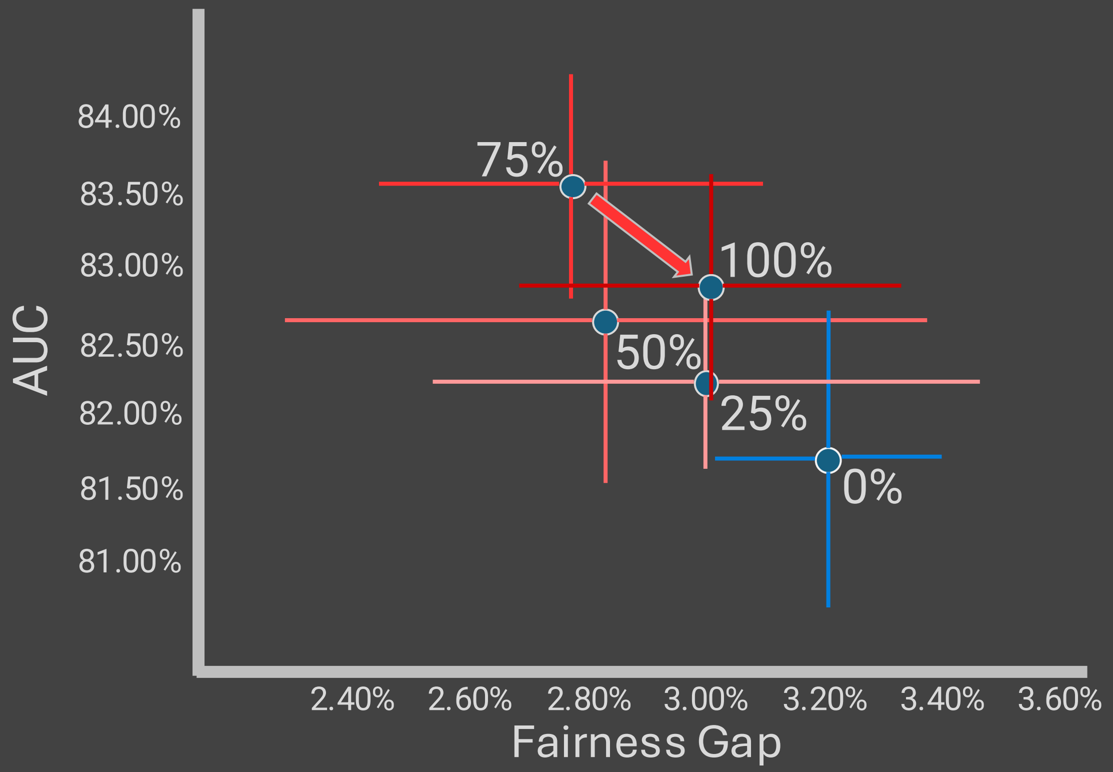

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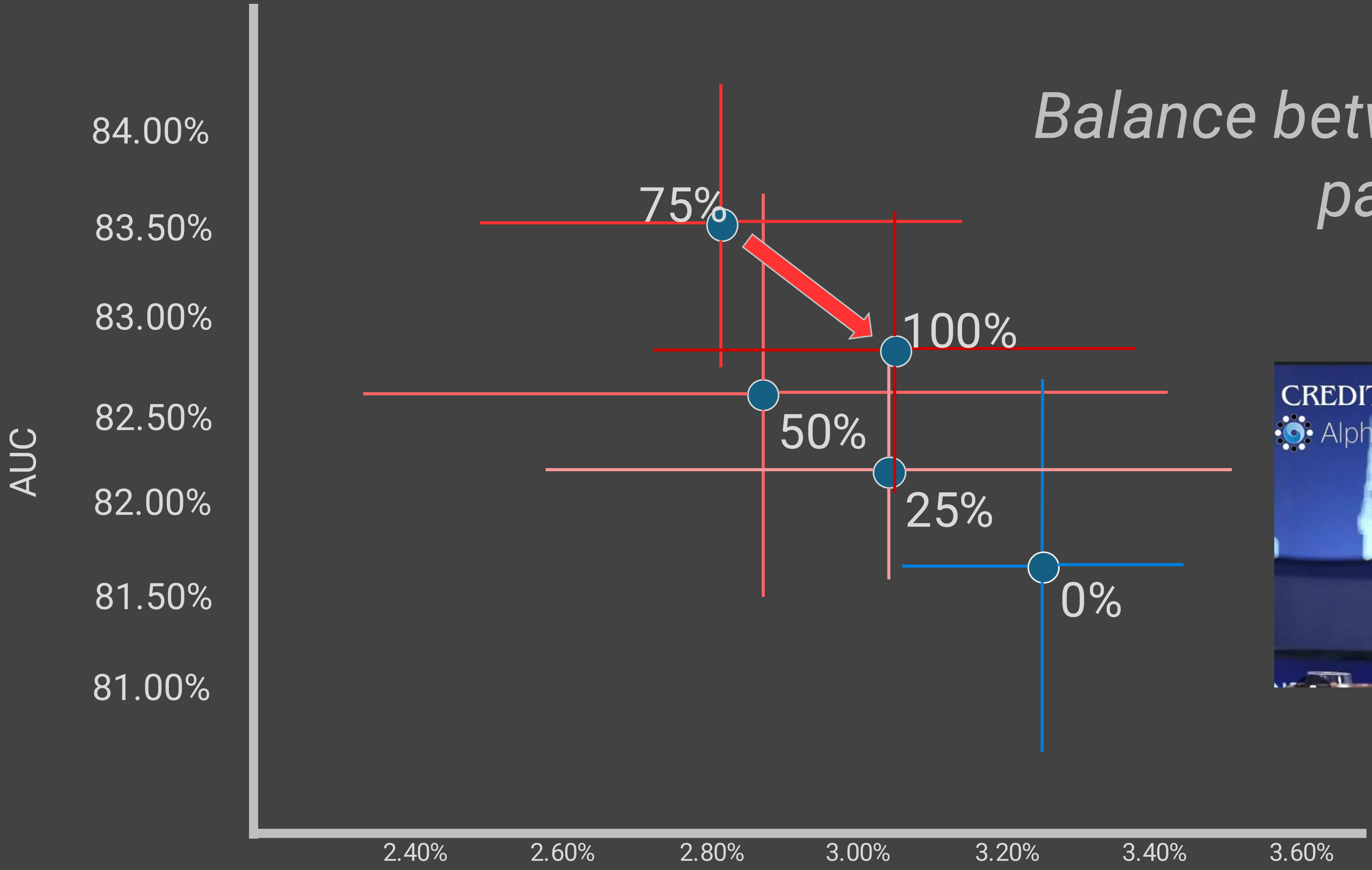

Balance between guidance and pattern discovery

Luo et al. Miccai'25

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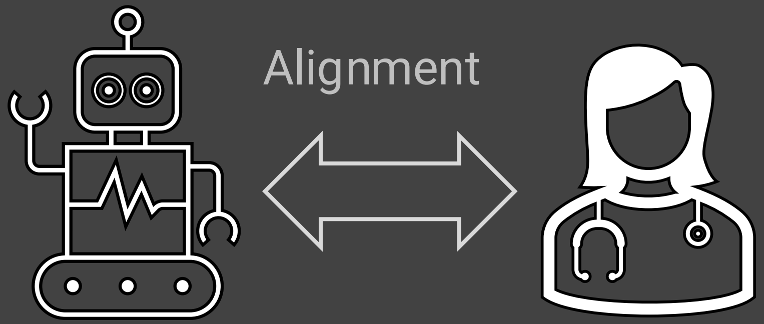
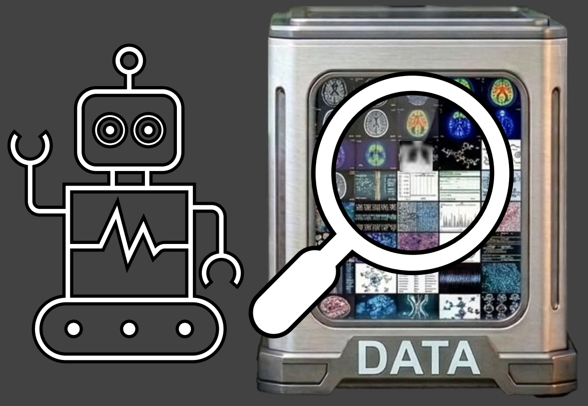
Balance between guidance and pattern discovery





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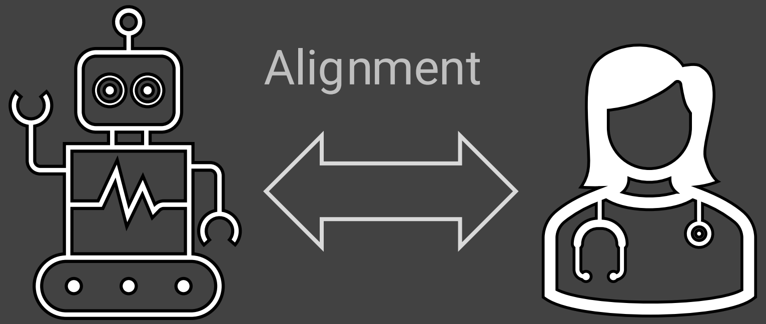
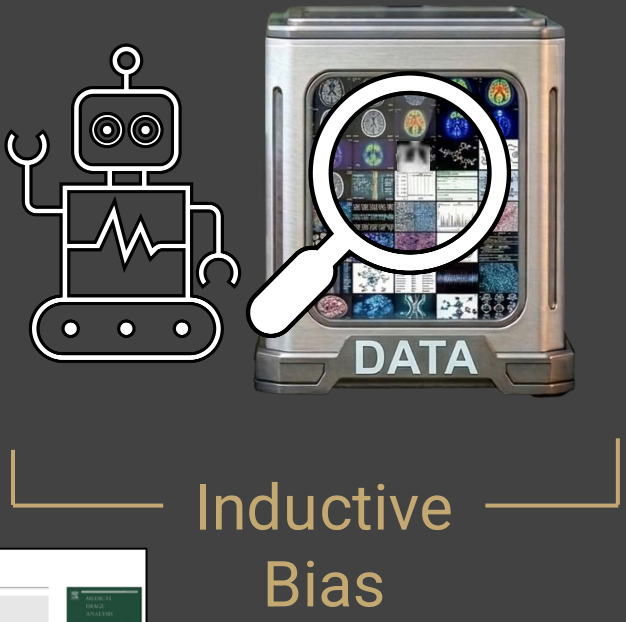
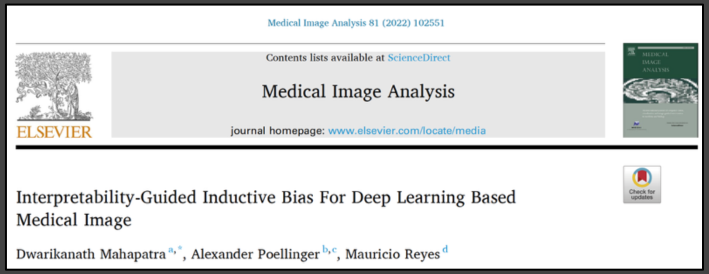
Balance between <u>guidance</u> and <u>pattern discovery</u>






Medical Image Analysis 81 (2022) 102551
Contents lists available at ScienceDirect
Medical Image Analysis
journal homepage: www.elsevier.com/locate/media

Interpretability-Guided Inductive Bias For Deep Learning Based Medical Image

Dwarikanath Mahapatra[a,*], Alexander Poellinger[b,c], Mauricio Reyes[d]

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Consolidation

Atelectasis

Cardiomegaly

Pleural Effusion

Edema

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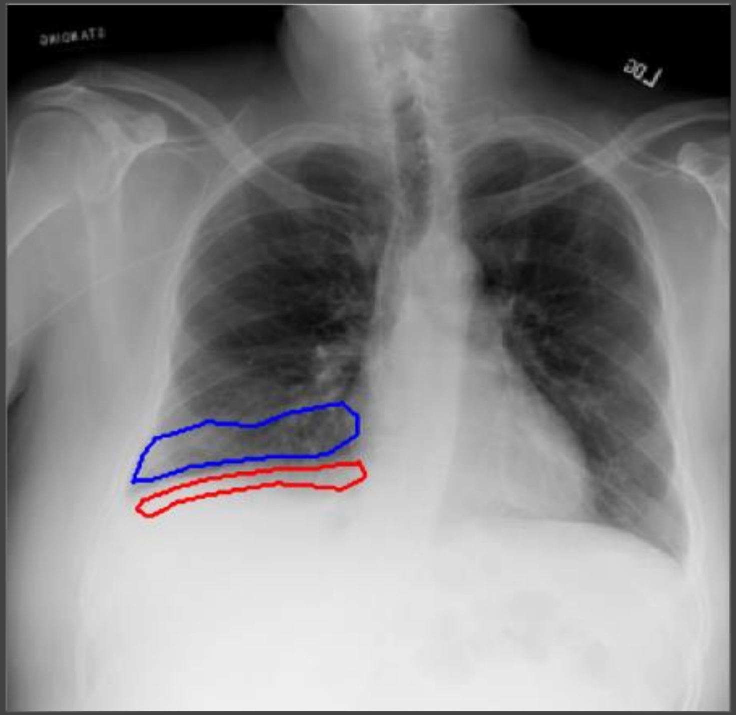

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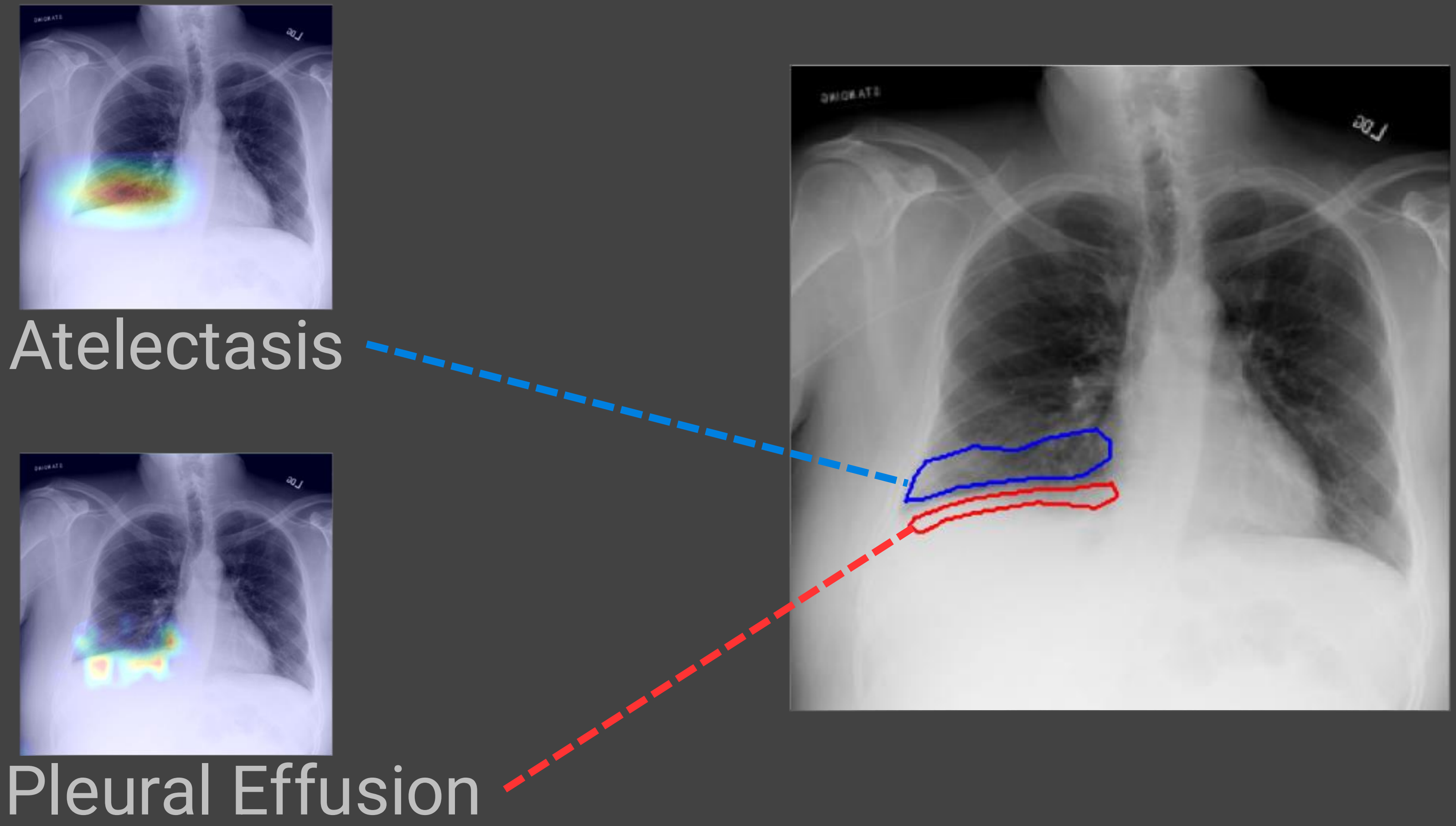

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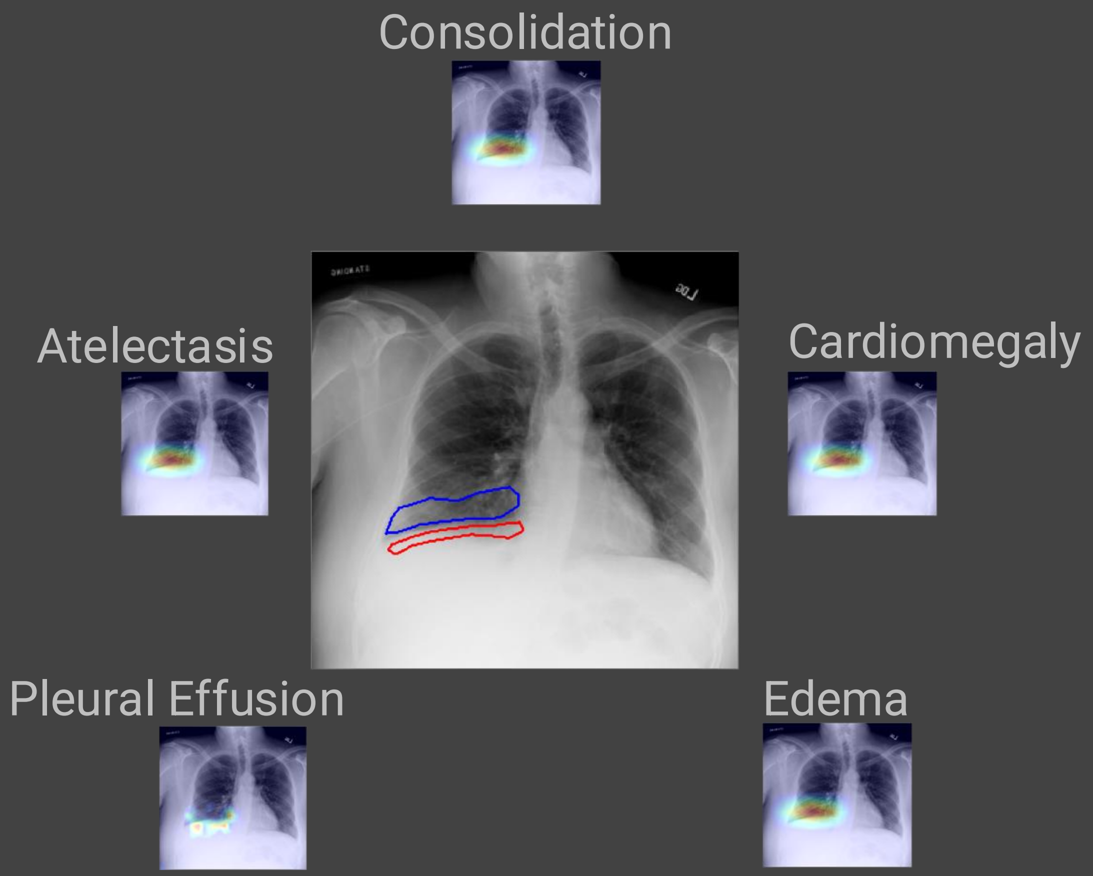

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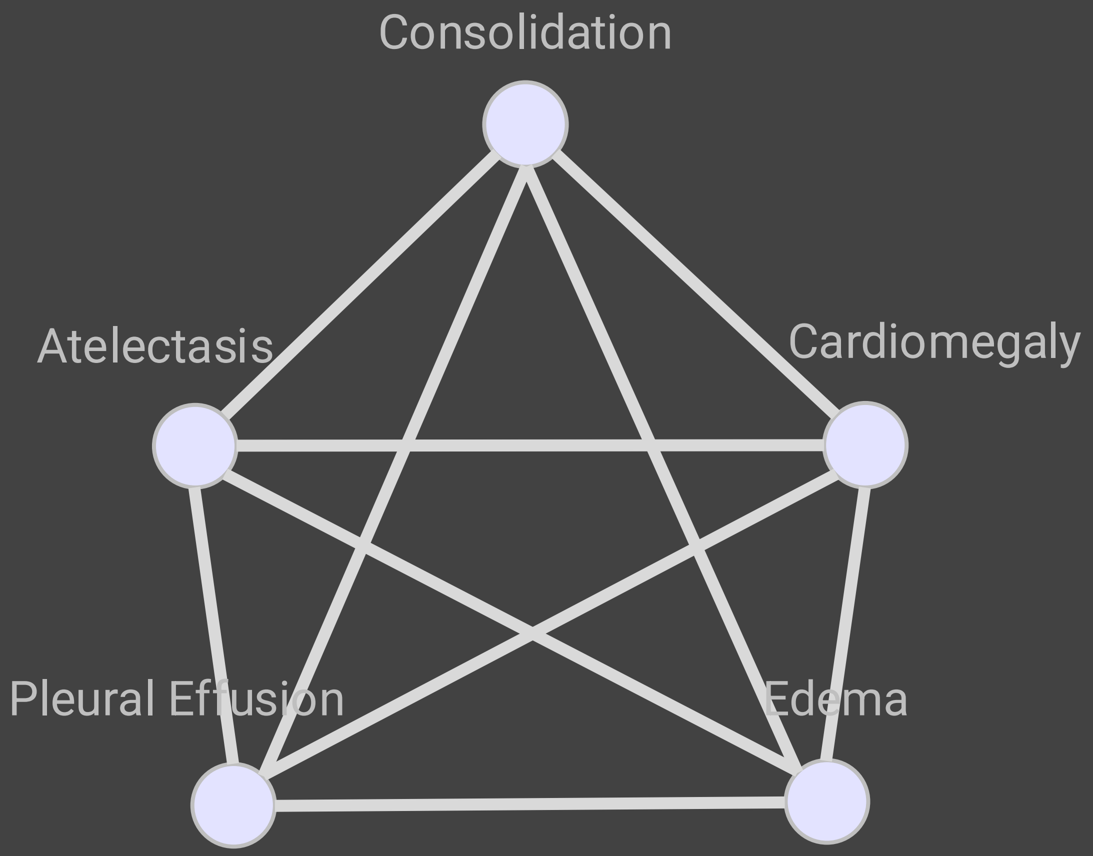

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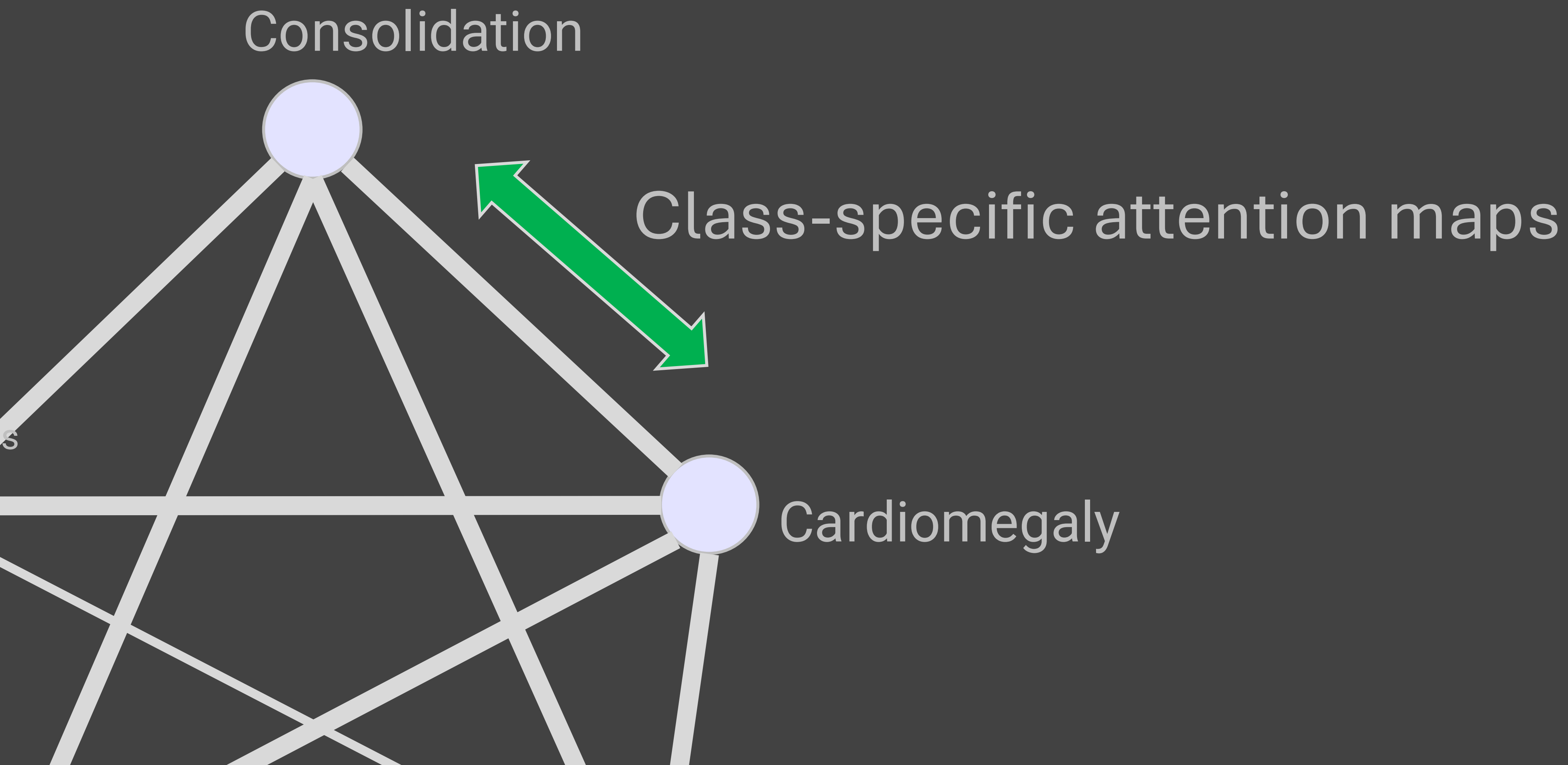
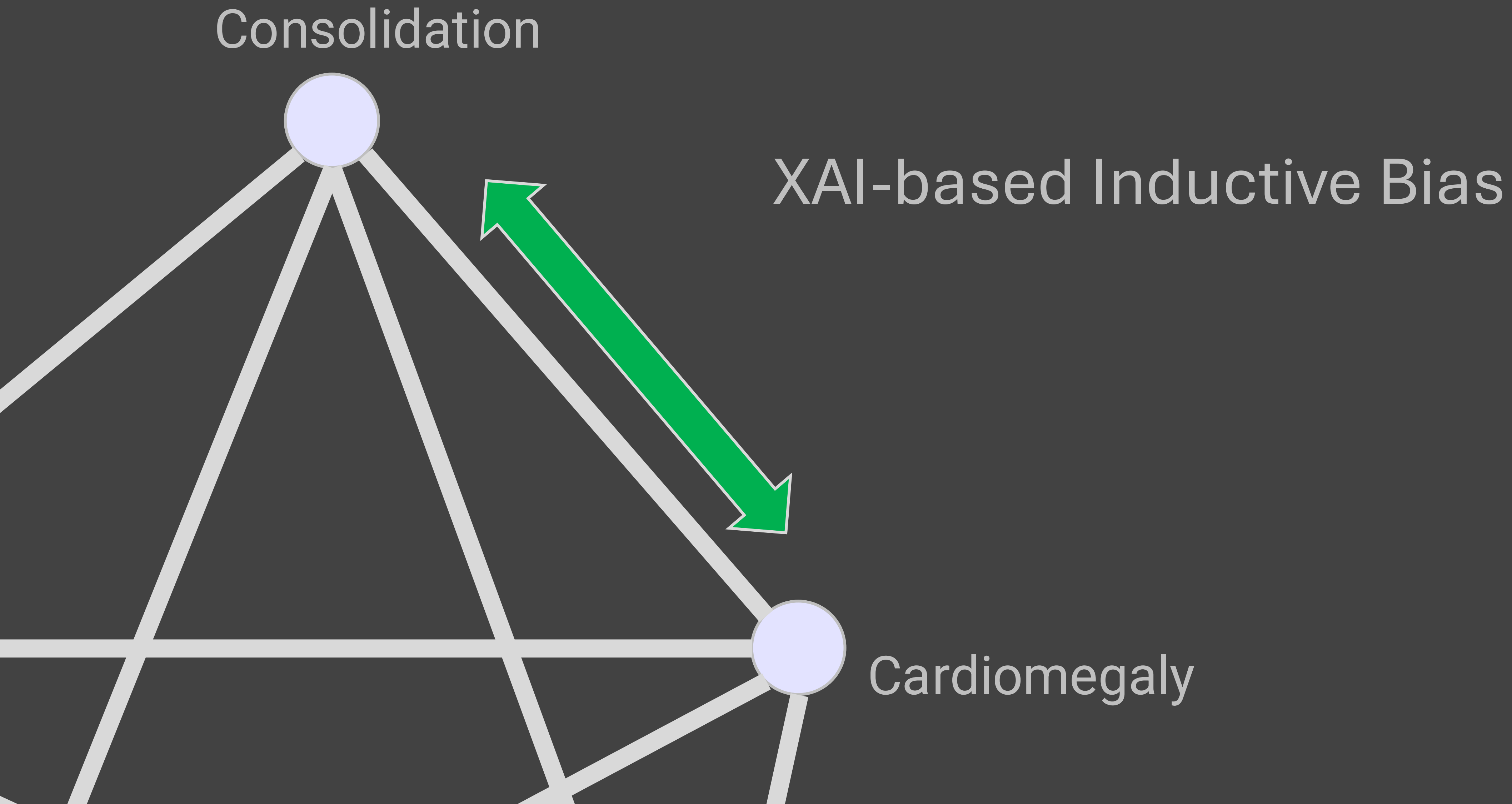
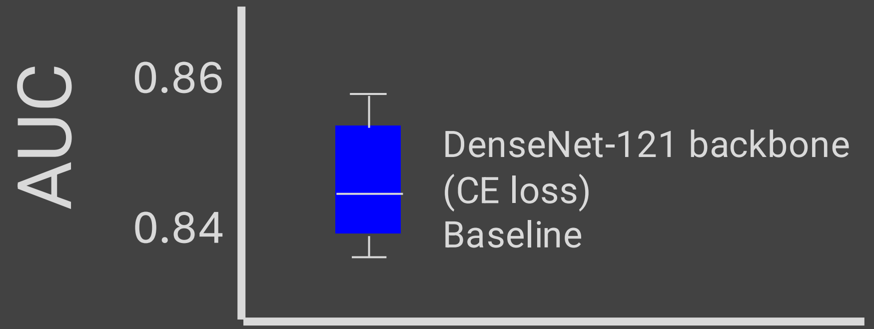


Reliable Clinical AI: Fair and Trustworthy



Reliable Clinical AI: Fair and Trustworthy

Saliency Inductive Bias: SIBNet

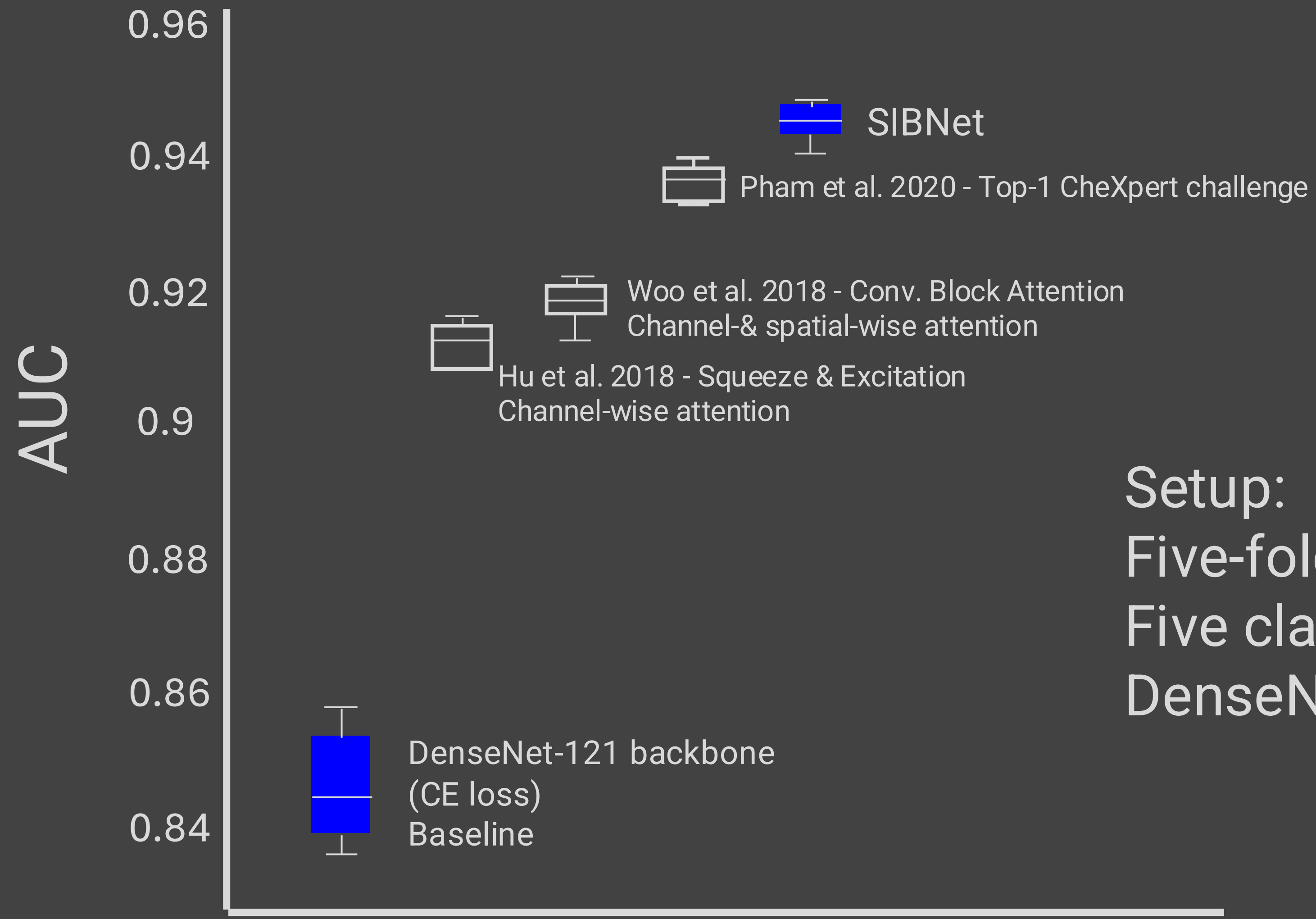


Setup:
Five-fold validation
Five classes
DenseNet-121 backbone

Mahapatra et al. MedIA 2022

Reliable Clinical AI: Fair and Trustworthy

Saliency Inductive Bias: SIBNet



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Mahapatra et al. MedIA 2022

Reliable Clinical AI: Fair and Trustworthy

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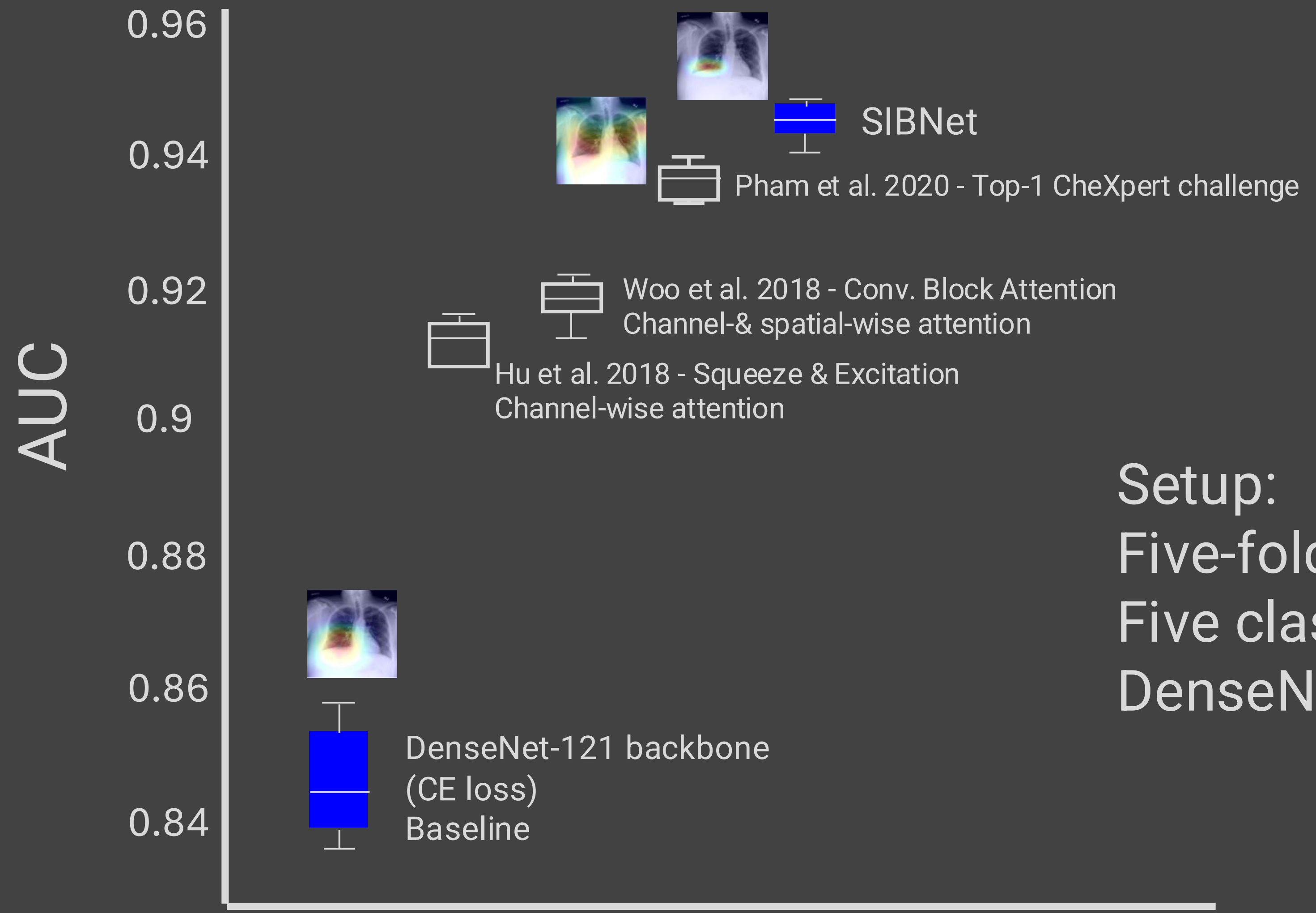

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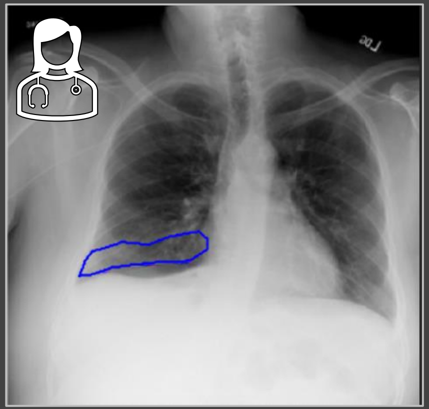
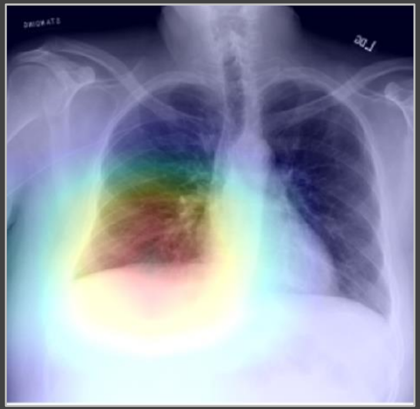
Mahapatra et al. MedIA 2022

Reliable Clinical AI: Fair and Trustworthy

Saliency Inductive Bias: SIBNet

Indirect Human-AI Alignment





DenseNet-121
(Baseline)

Pham et al. 2020

SIBNet

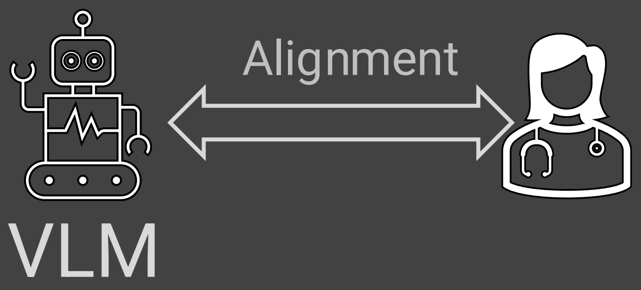
Mahapatra et al. MedIA 2022

Reliable Clinical AI: Fair and Trustworthy

Combining Human-AI alignment and XAI-based Inductive Bias

Reliable Clinical AI: Fair and Trustworthy

Combining Human-AI alignment and XAI-based Inductive Bias



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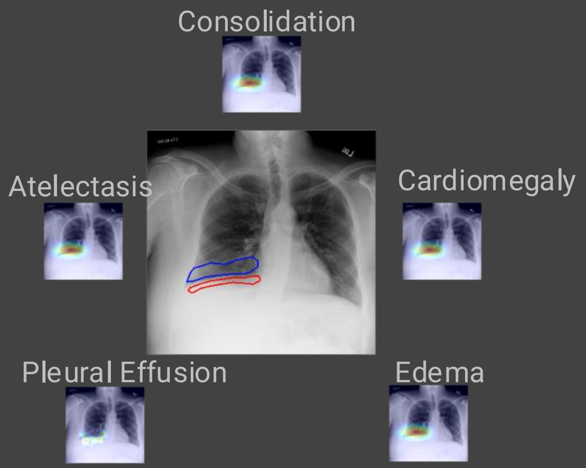
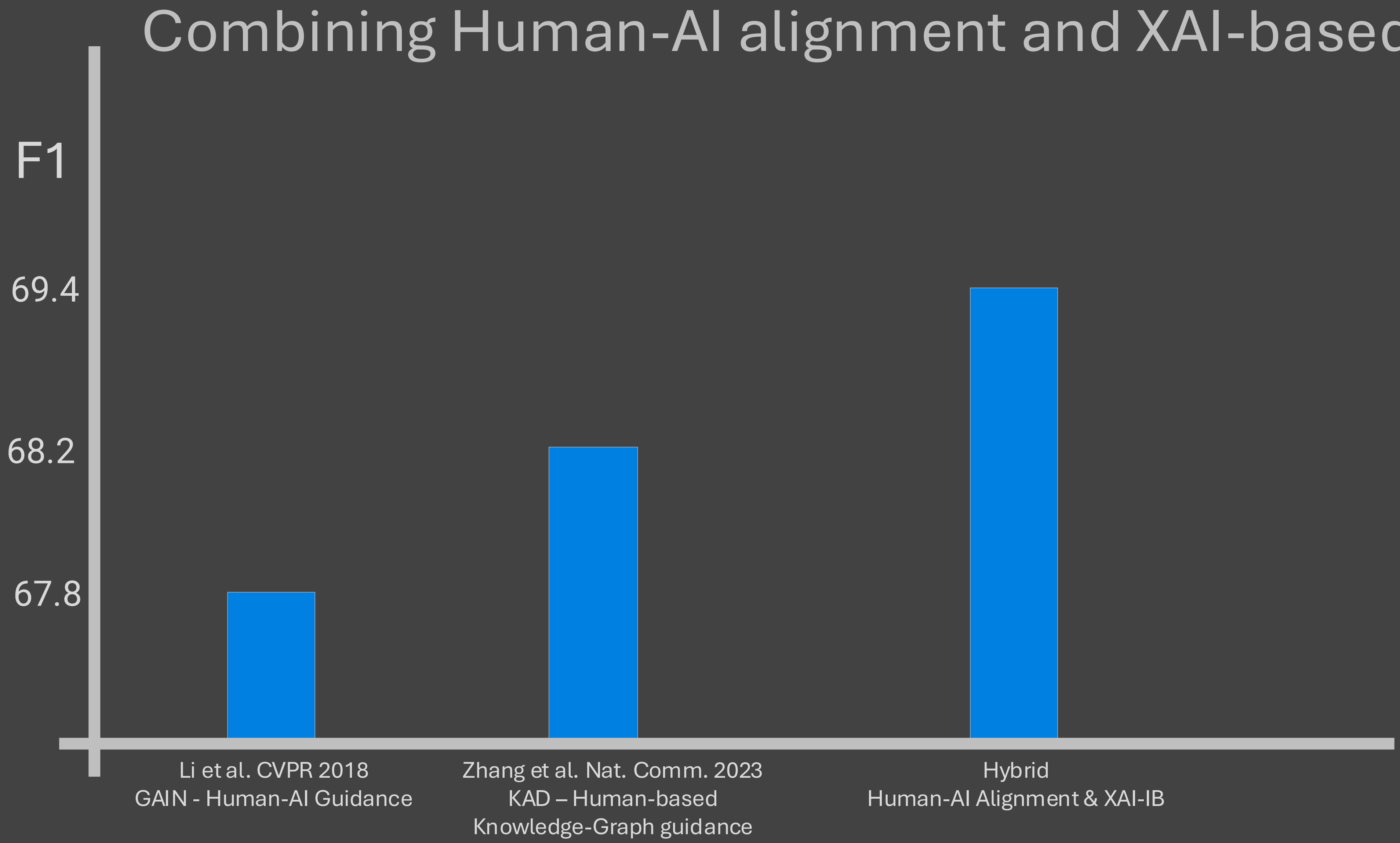


Shu et al. Miccai-iMIMIC 2025

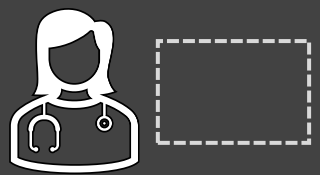
Reliable Clinical AI: Fair and Trustworthy

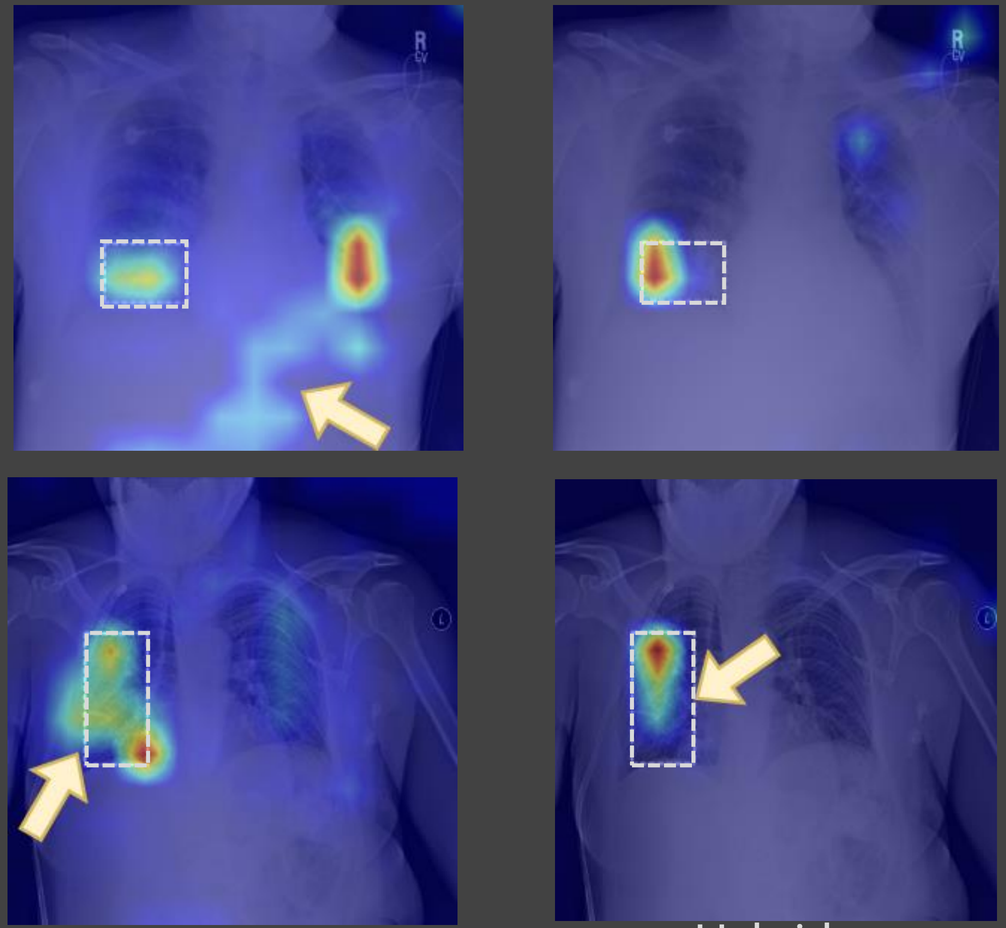


Shu et al. Miccai-iMIMIC 2025

Reliable Clinical AI: Fair and Trustworthy

Combining Human-AI alignment and XAI-based Inductive Bias



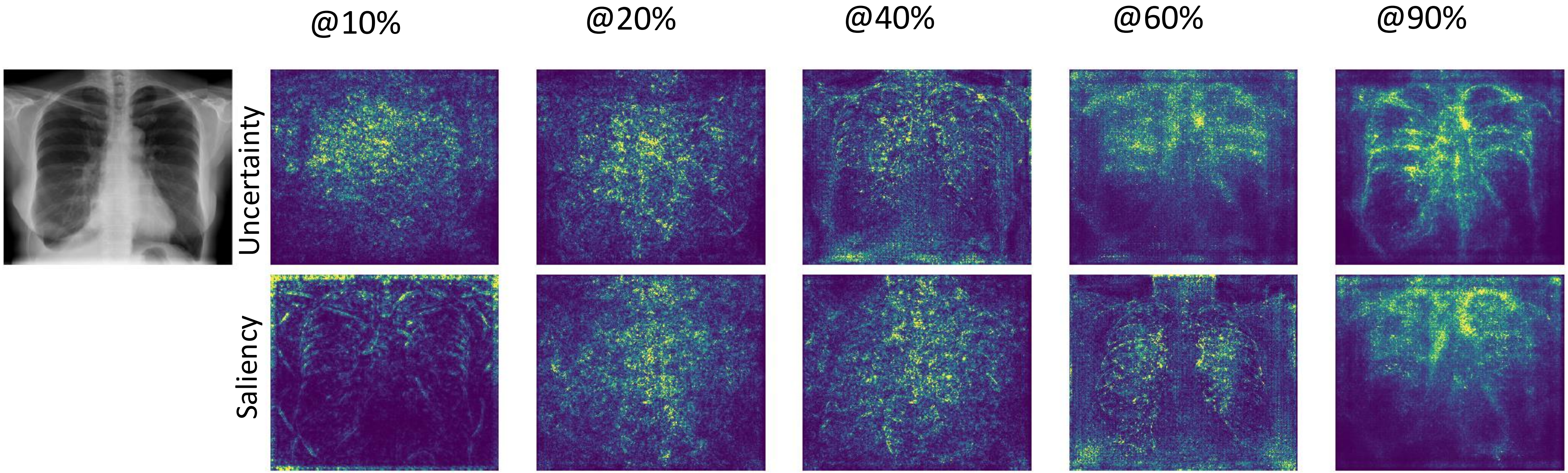


Zhang et al. Nat. Comm. 2023
KAD – Human-based
Knowledge-Graph guidance

Hybrid
Human-AI & XAI-IB

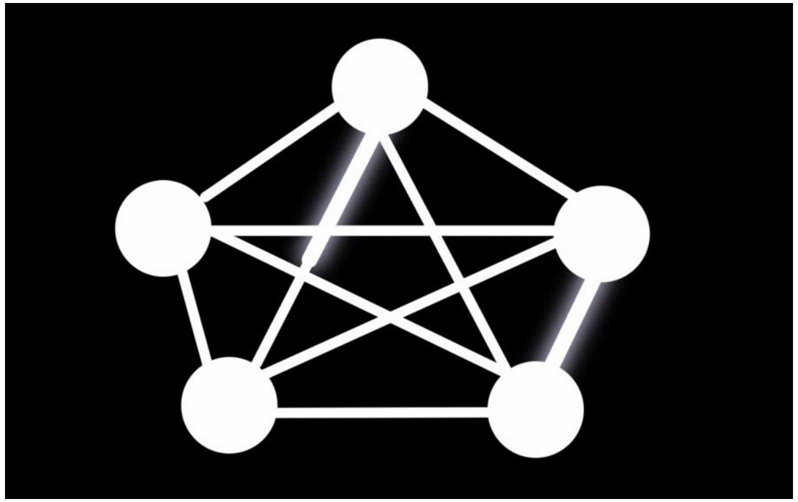
Shu et al. Miccai-iMIMIC 2025

Saliency maps evolve during training



Key message: Saliency maps can be seen/used as a fingerprint of model's behavior to inputs

Graph Node-based Interpretability Guided for Active Learning?

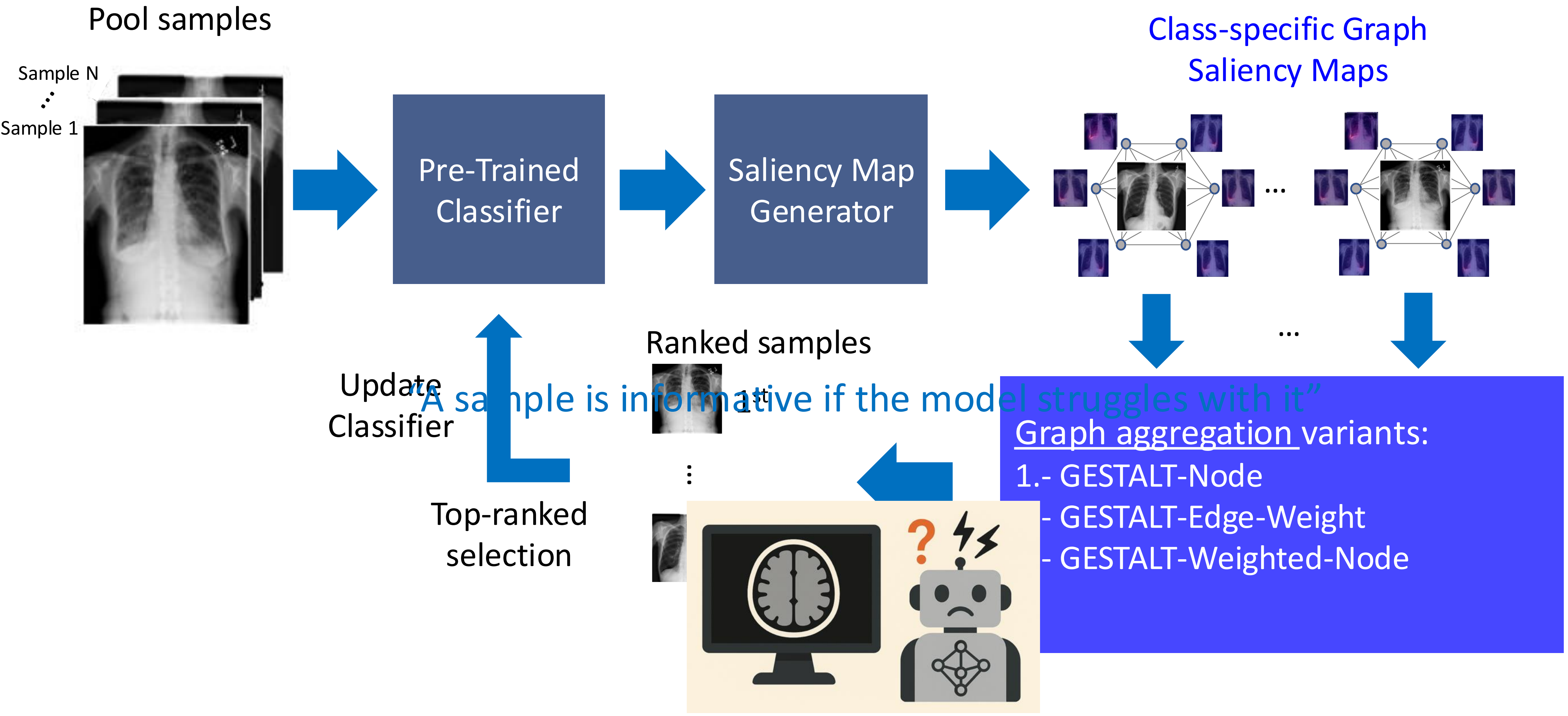


IEEE TRANSACTIONS ON MEDICAL IMAGING, VOL. 42, NO. 3, MARCH 2023

Graph Node Based Interpretability Guided Sample Selection for Active Learning

Dwarikanath Mahapatra, Alexander Poellinger, and Mauricio Reyes

Graph Node-Based Interpretability Guided Sample Selection (GESTALT)



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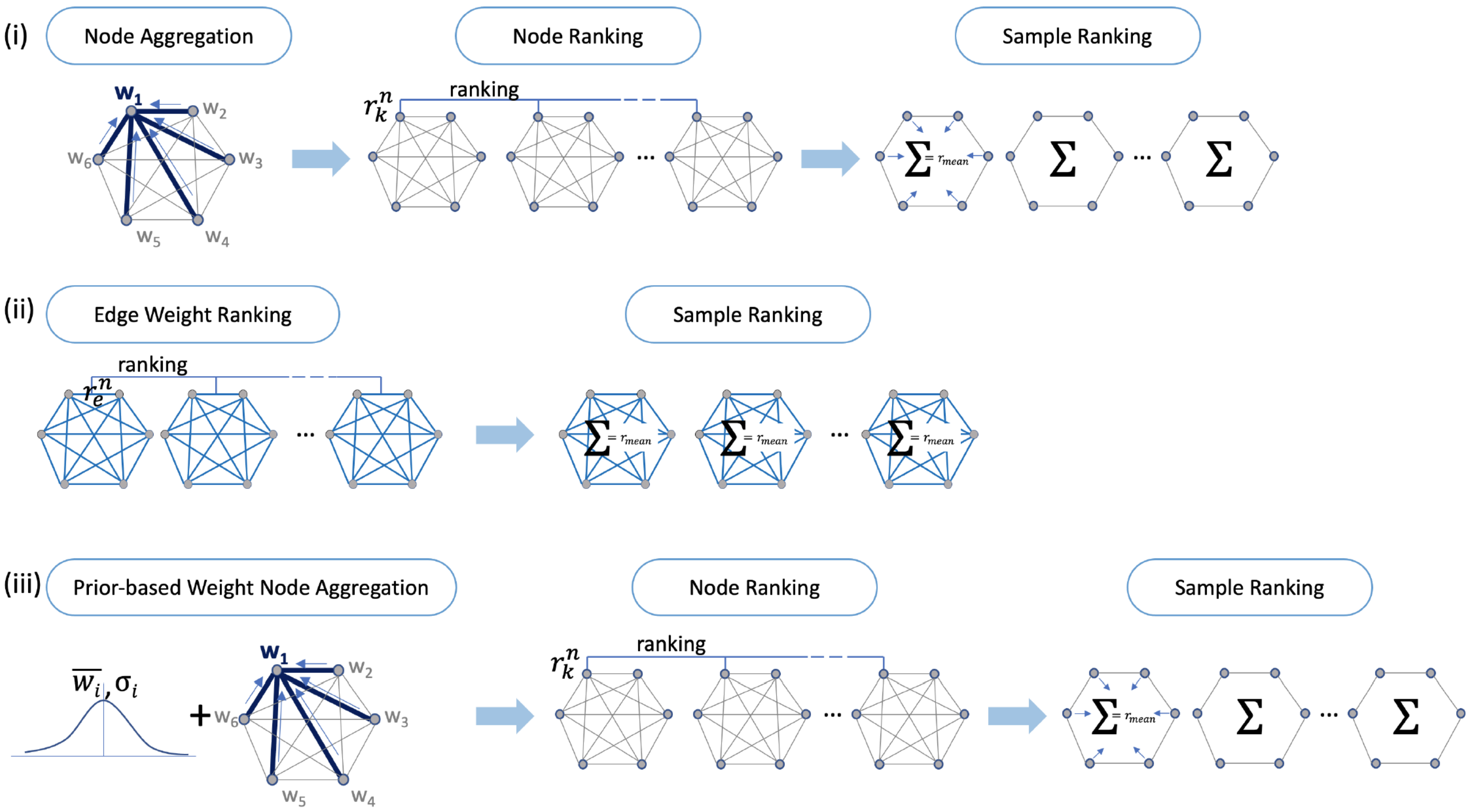
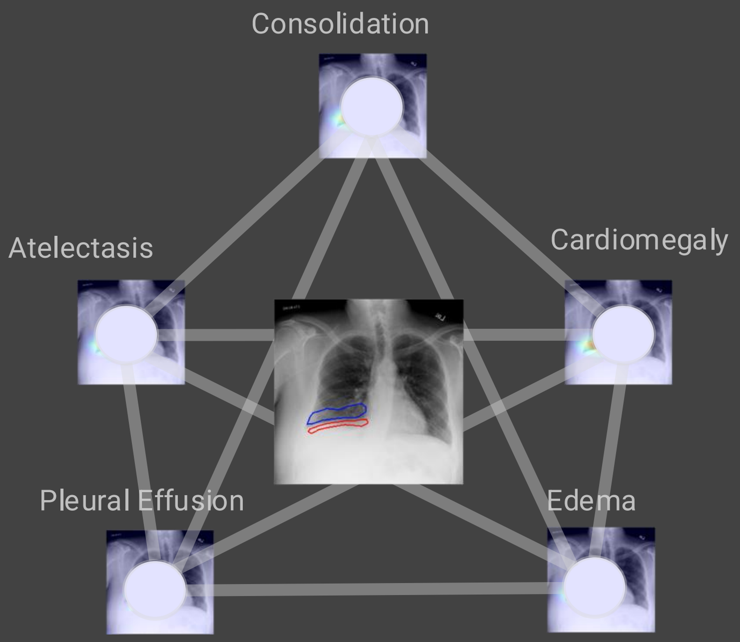


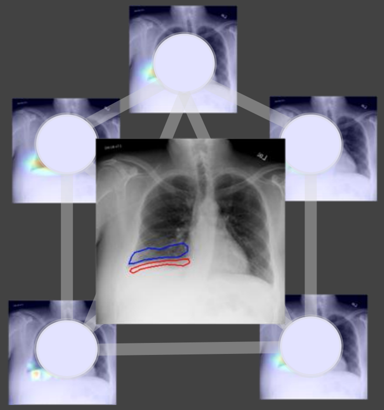
XAI-boosted Active Learning

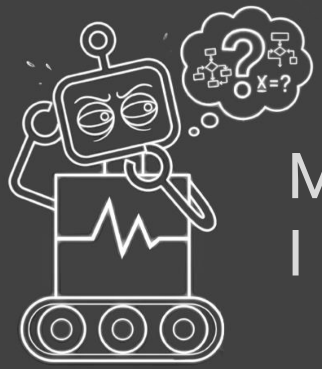


Mmh, <u>easy</u> sample ...

Mahapatra et al. MedIA 2024
Mahapatra et al. MedIA 2024b
Shu et al. Miccai-iMIMIC 2024
Mahapatra et al. TMI 2022

XAI-boosted Active Learning





Mmh, a <u>hard</u> (a.k.a informative) sample...
I can learn from it!

Mahapatra et al. MedIA 2024
Mahapatra et al. MedIA 2024b
Shu et al. Miccai-iMIMIC 2024
Mahapatra et al. TMI 2022

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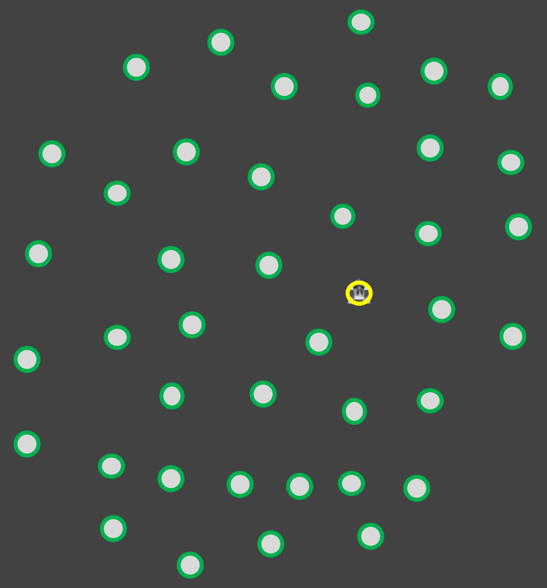

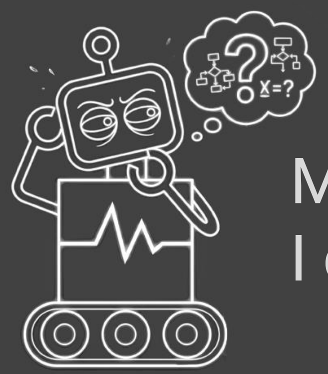

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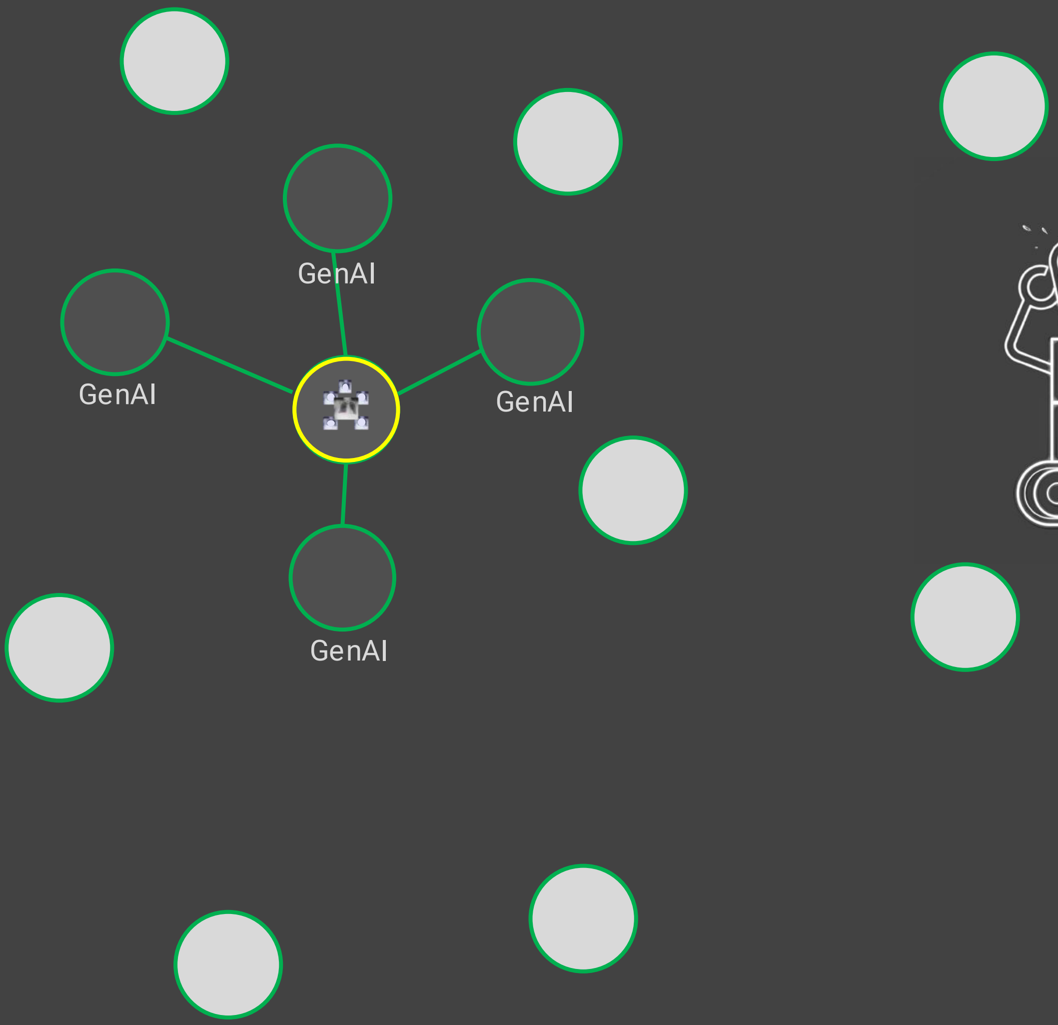
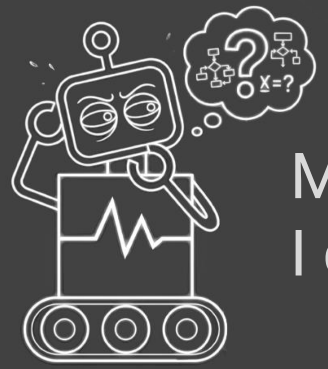
XAI-boosted Active Learning





Mmh, a <u>hard</u> (a.k.a informative) sample...
I can learn from it!

Mahapatra et al. MedIA 2024
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Shu et al. Miccai-iMIMIC 2024
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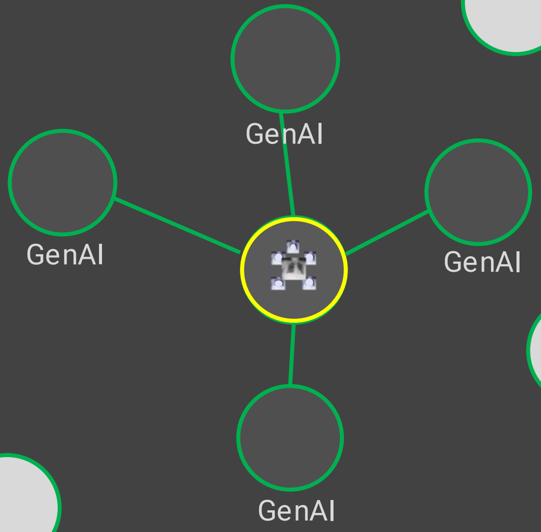

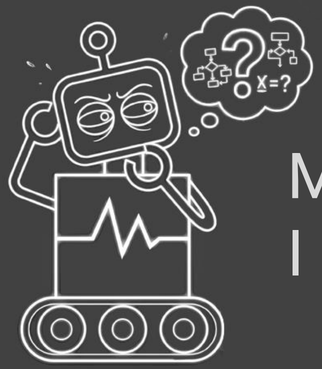

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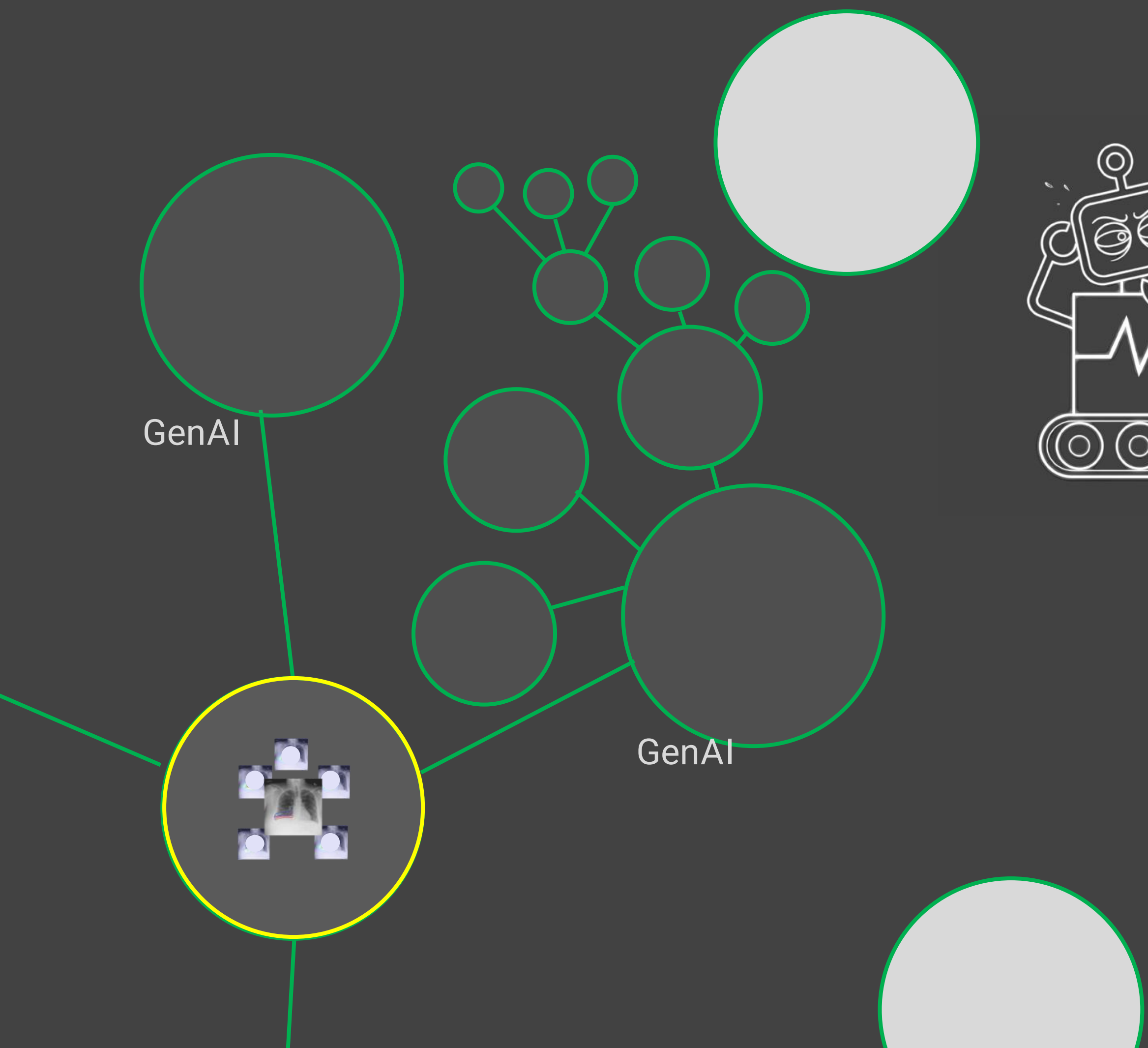

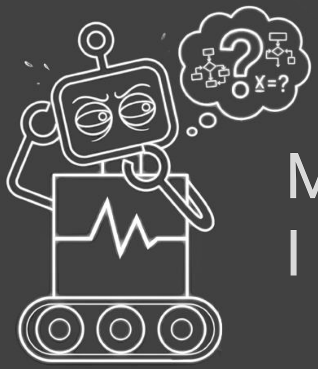

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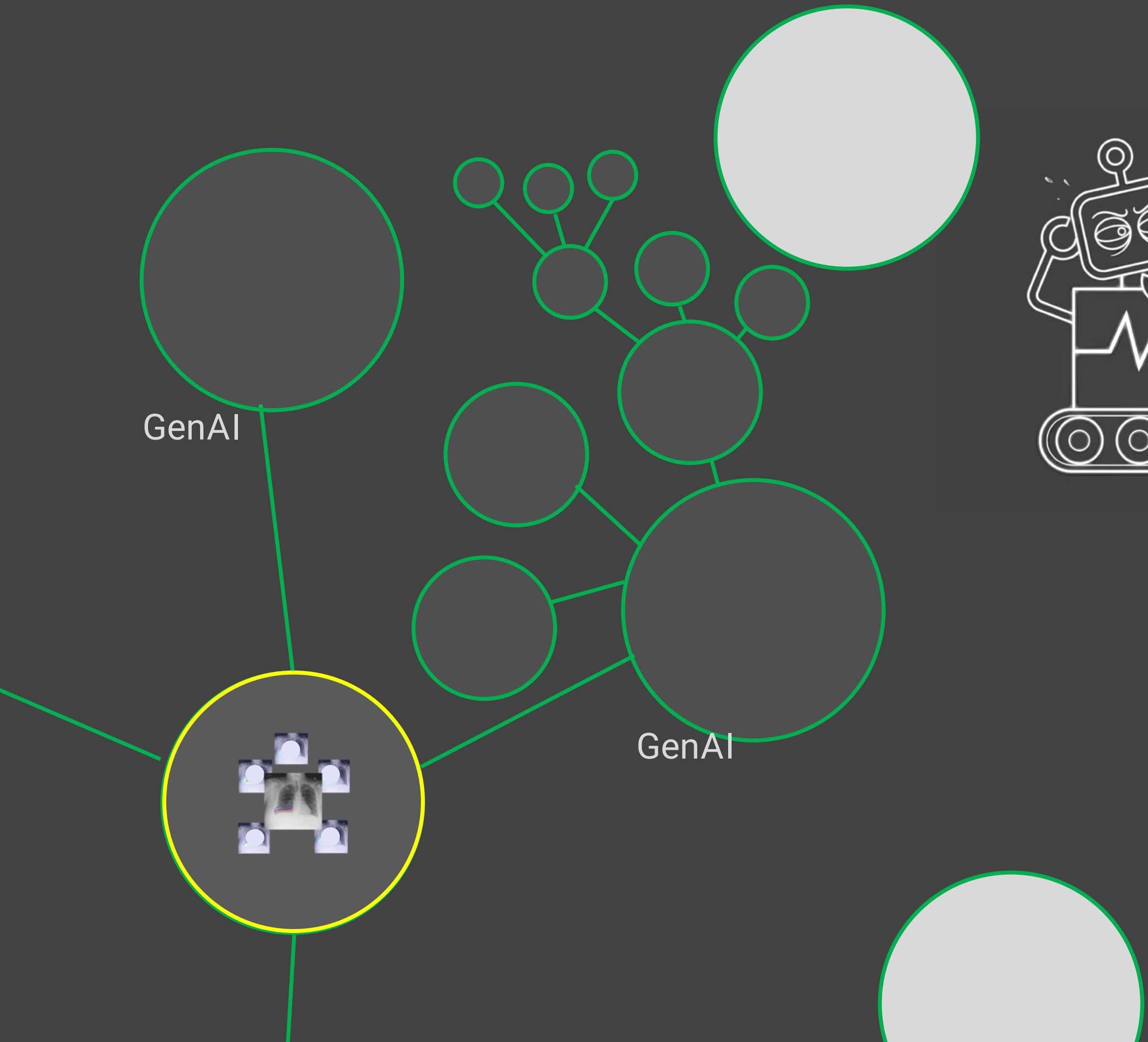

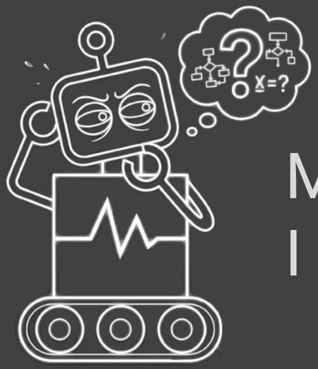

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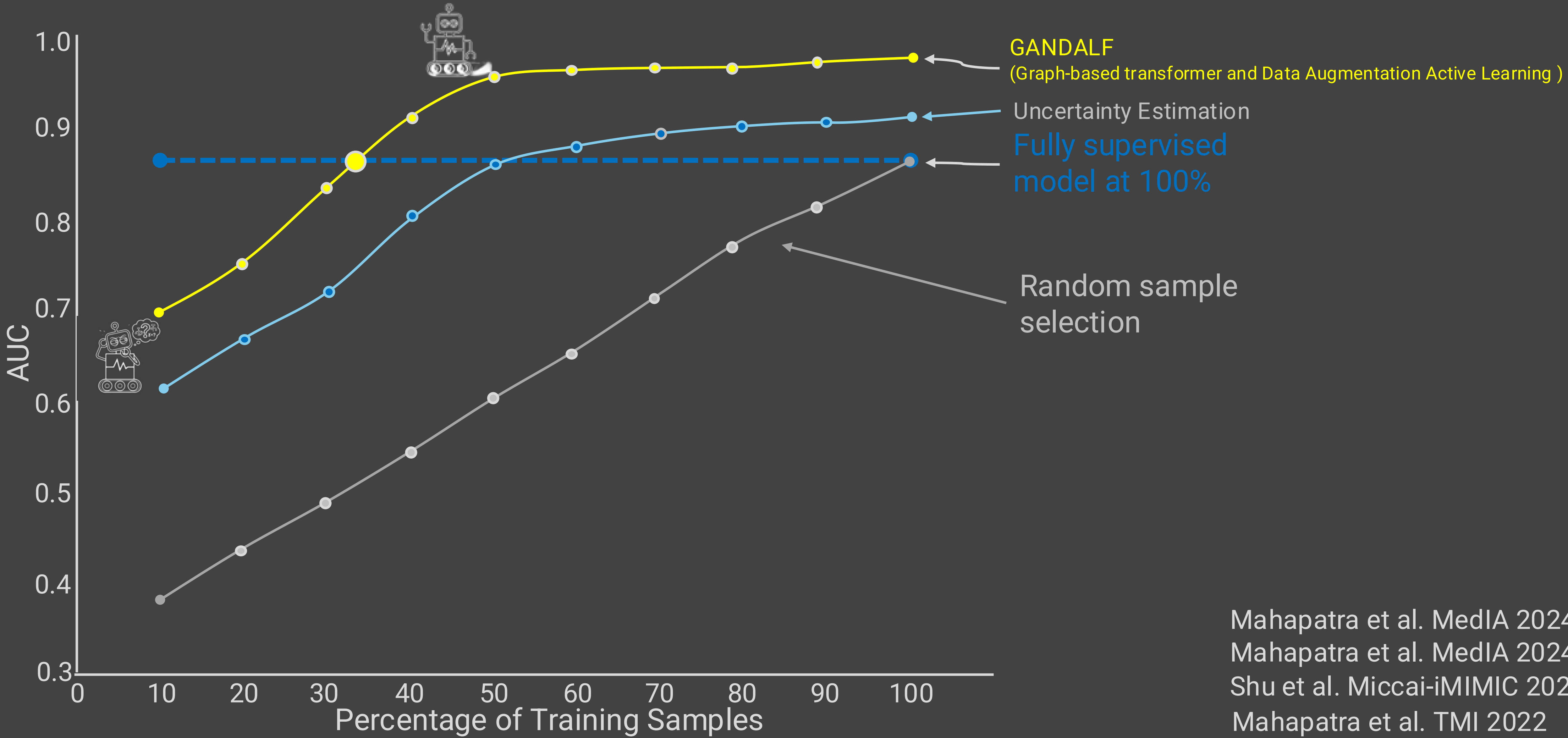

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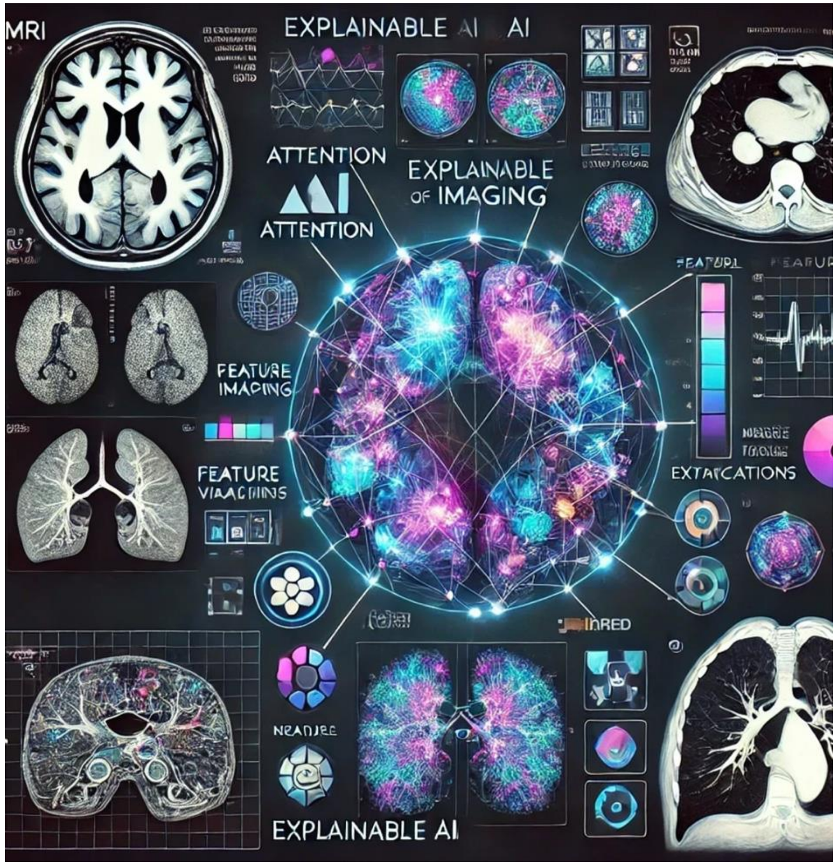
Let’s wrap-up

- Interpretability of AI systems is propelled by new findings from the DL community to safely translate this technology to (medical) applications.
- The goal of interpretability is to have enough information for the task at hand. In medicine: Patient-safety is first.
- Several types of XAI methods
- Current challenges: XAI needs to be multimodal and longitudinal to match clinical workflows.
- XAI in the era of Foundation Models: Bigger Black Box?



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